

Two-Timescale Secure Transmission for Movable Antenna-Enabled Multiuser Systems With Statistical Eavesdropper CSI

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Abstract—Movable antennas (MAs) can reshape the spatial correlation between the channels of legitimate users (LUs) and an eavesdropper (Eve), which provides a new degree of freedom for physical layer security. However, existing MA-aided secure designs rely on instantaneous channel state information (CSI), which is hindered by the antenna movement delay, the CSI acquisition overhead over the movable region, and the unavailability of the instantaneous CSI of a passive Eve. To overcome these limitations, we propose a two-timescale secure transmission framework for an MA-enabled multiuser downlink system, where the antenna positions and the power allocation between the confidential signals and artificial noise (AN) are optimized based on statistical CSI, whereas zero-forcing precoding and null-space AN are designed based on the instantaneous CSI of the LUs only. We prove that the information leakage and the AN received by Eve are conditionally independent given the channels of the LUs, and derive a closed-form approximation of the ergodic wiretap rate, which shows that the antenna positions affect the ergodic secrecy sum rate (ESSR) only through three geometric quantities. Then, we develop an alternating optimization algorithm, where the power allocation admits a secrecy water-filling structure, and each antenna position is updated by a global search followed by majorization–minimization refinement. Simulation results show that, at a transmit signal-to-noise ratio of 20 dB, the proposed design improves the ESSR by 14.8% and 31.2% over the rate-oriented two-timescale MA design and a fixed-position array, respectively, and reduces the required AN power.

Index Terms—Movable antenna, physical layer security, artificial noise, two-timescale design, statistical CSI, ergodic secrecy rate.

I. INTRODUCTION

PHYSICAL layer security (PLS) exploits the randomness and the spatial characteristics of wireless channels to protect confidential messages against eavesdropping, which complements upper-layer cryptography [1]–[3]. In multi-antenna systems, secure precoding and artificial noise (AN) are two main enablers of PLS [4], [5], whose performance is limited by how well the legitimate users (LUs) and the eavesdropper (Eve) can be spatially separated. When Eve is located in a direction close to that of an LU and the line-of-sight (LoS) propagation is dominant, their channels are highly correlated, and a conventional fixed-position antenna (FPA) array can hardly steer the information-bearing beams away from Eve without sacrificing the beamforming gain at the LU.

Movable antennas (MAs), also known as fluid antennas, have recently emerged as a promising technology to overcome this limitation [6]–[8]. By adjusting the antenna positions

within a given region, MAs reshape the channel correlation among different receivers without additional radio-frequency (RF) chains [9]–[11], which can be exploited to decorrelate the channels of the LUs from that of Eve. The authors of [12] jointly optimized the transmit beamforming and the positions of an MA array to maximize the secrecy rate with perfect channel state information (CSI), and the statistical CSI of the eavesdroppers was exploited in [13] when their instantaneous CSI is unavailable. More recently, a deterministic equivalent of the ergodic secrecy rate of a multiple-input multiple-output multiple-eavesdropper-antenna MA system with partial CSI of Eve was derived in [14], and a sensing-assisted secure design was developed in [15], where the MA positions enhance both the estimation of the direction of Eve and the secrecy rate.

Despite these advances, the above works optimize the antenna positions based on the instantaneous CSI of at least the LUs, which requires the antennas to be repositioned in each channel coherence interval (CI) under fast fading. Such single-timescale designs are difficult to implement, since the mechanical movement incurs non-negligible delay and energy consumption, the CSI is required over the entire movable region, which, under Rician fading, can only be acquired by traversing the region [16], [17], and the instantaneous CSI of a passive Eve is generally unavailable. To reduce the overhead, two-timescale MA designs have been proposed [16], [17], following the paradigm developed for intelligent reflecting surfaces [18], where the antenna positions are optimized based on the statistical CSI, and the beamforming is designed based on the instantaneous CSI of the resulting effective channels. However, these works focused on the ergodic sum rate, and the two-timescale design of MA-enabled secure transmission remains unexplored.

Extending the two-timescale MA design to secure transmission is not straightforward. First, the ergodic wiretap rate of Eve is determined by the random precoding directions and AN subspace induced by the instantaneous CSI of the LUs, whereas the existing ergodic-rate approximations for zero-forcing (ZF) precoding [16], [17], [19] only characterize the signal power received by the intended users. Second, in the rate-oriented design, the antenna positions improve the ergodic rate only by decorrelating the users, and thus bring no gain for a single user [16], whereas in secure transmission they also determine the information leakage and the AN received by Eve. Third, the power allocation between the confidential signals and the AN is coupled with the antenna positions. To address these challenges, this paper makes the following main

contributions.

- We propose a two-timescale secure transmission framework for MA-enabled multiuser systems, where the antenna positions ($\mathbf{t}$) and the long-term power allocation among the confidential signals and AN ($\mathbf{p}, \mathbf{q}$) are optimized based on the statistical CSI of the LUs and Eve, whereas ZF precoding and null-space AN are designed based on the instantaneous CSI of the LUs only, which requires neither the instantaneous CSI of Eve nor frequent antenna movement.
- We prove that the information leakage and the AN received by Eve are conditionally independent given the channels of the LUs, and derive a closed-form approximation of the ergodic wiretap rate, which is exact in the non-LoS (NLoS)-dominant regime and asymptotically exact in the LoS-dominant regime. The resulting approximate ergodic secrecy sum rate (ESSR) depends on $\mathbf{t}$ only through three geometric quantities, which reveals that the MAs enhance secrecy by decorrelating the LUs, reducing the information leakage, and increasing the AN received by Eve.
- We develop an alternating optimization (AO) algorithm to maximize the approximate ESSR, where the optimal power allocation for given positions admits a secrecy water-filling structure, and each antenna position is updated by a global search over the movable region, which exploits a rank-one structure of the objective function, followed by majorization-minimization (MM) refinement. The algorithm converges monotonically within a few iterations, and its complexity does not depend on the number of channel realizations.
- Simulation results show that, at a transmit signal-to-noise ratio (SNR) of 20 dB, the proposed design improves the ESSR by 14.8% and 31.2% over the rate-oriented two-timescale MA design [16] and an FPA array, respectively, and by 20.4% over antenna selection (AS) from twice as many antennas, while requiring considerably less AN power. The gain persists for a single LU, against an Eve that treats the inter-user interference as noise, and under moderate errors in the direction of Eve.

The rest of this paper is organized as follows. Section II presents the system model and problem formulation. Section III derives the approximate ESSR. Section IV develops the optimization algorithm. Section V provides simulation results, and Section VI concludes the paper.

Notations: Scalars, vectors, and matrices are denoted by italic, bold lower-case, and bold upper-case letters, respectively. $(\cdot)^T$, $(\cdot)^*$, and $(\cdot)^H$ denote the transpose, conjugate, and conjugate transpose, and $|\cdot|$, $\|\cdot\|$, and $\text{tr}(\cdot)$ the modulus, Euclidean norm, and trace, respectively. $[\mathbf{x}]_i$ and $[\mathbf{X}]_{i,j}$ denote the i -th entry of $\mathbf{x}$ and the (i,j) -th entry of $\mathbf{X}$, $\text{diag}(x_1, \dots, x_M)$ a diagonal matrix, and $\mathbf{X}^{1/2}$ the entrywise square root of a nonnegative diagonal matrix $\mathbf{X}$. $\mathbf{I}_N$ denotes the $N \times N$ identity matrix, $\mathbf{e}_m$ the m -th column of an identity matrix, and $\mathcal{CN}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ the circularly symmetric complex Gaussian distribution. $\mathbf{X} \succeq \mathbf{0}$ ($\mathbf{X} \succ \mathbf{0}$) means that $\mathbf{X}$ is positive semidefinite (definite). $j \triangleq \sqrt{-1}$, $\mathbb{E}\{\cdot\}$ and $\text{Var}\{\cdot\}$ denote the expectation

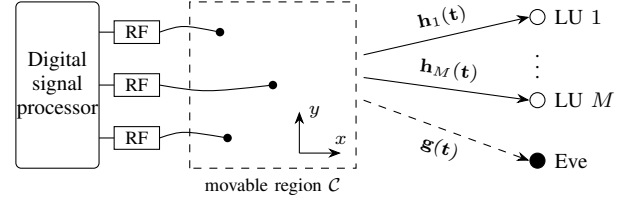


Fig. 1. An MA-enabled multiuser downlink system in the presence of a passive eavesdropper.

and variance, $\text{Re}\{\cdot\}$ the real part, $[x]^+ \triangleq \max\{x, 0\}$, and $|\mathcal{S}|$ the cardinality of a set $\mathcal{S}$.

II. SYSTEM MODEL AND PROBLEM FORMULATION

A. MA-Enabled Secure Multiuser Downlink System

As illustrated in Fig. 1, we consider the downlink of a multiuser system, where a base station (BS) equipped with N MAs serves M single-antenna LUs in the presence of a single-antenna passive Eve. The LUs and the MAs are indexed by $m \in \mathcal{M} \triangleq \{1, \dots, M\}$ and $n \in \mathcal{N} \triangleq \{1, \dots, N\}$, respectively, where $N \geq M + 1$ so that AN can be transmitted in the null space of the channels of the LUs. The MAs are connected to the RF chains via flexible cables and can move within a two-dimensional (2D) square region $\mathcal{C} \triangleq [-A/2, A/2] \times [-A/2, A/2]$ of side length A [6], [7]. In the local Cartesian coordinate system of $\mathcal{C}$, the position of the n -th MA is denoted by $\mathbf{t}_n \triangleq [x_n, y_n]^T \in \mathcal{C}$, and the positions of all the MAs are collected in $\mathbf{t} \triangleq [\mathbf{t}_1^T, \dots, \mathbf{t}_N^T]^T \in \mathbb{R}^{2N \times 1}$. To avoid the coupling between adjacent antennas, a minimum inter-MA distance $D_{\min}$ is required, i.e., $\|\mathbf{t}_n - \mathbf{t}_i\| \geq D_{\min}$, $\forall n \neq i$.

Since the size of $\mathcal{C}$ is much smaller than the propagation distance, moving the MAs changes only the phase of the LoS path, whereas its angle of departure (AoD) and amplitude remain unchanged [7]. Let $\theta_m \in [-\pi/2, \pi/2]$ and $\phi_m \in [-\pi/2, \pi/2]$ denote the elevation and azimuth AoDs of the LoS path from the BS to LU m , respectively, and define the wave vector $\mathbf{a}_m \triangleq [\cos \theta_m \sin \phi_m, \sin \theta_m]^T$. Then, the propagation distance difference of the LoS path between the n -th MA and the origin of $\mathcal{C}$ is $\mathbf{t}_n^T \mathbf{a}_m$, and the LoS component of the channel from the BS to LU m is given by [16]

$$\bar{\mathbf{h}}_m(\mathbf{t}) \triangleq \left[e^{j \frac{2\pi}{\lambda} \mathbf{t}_1^T \mathbf{a}_m}, \dots, e^{j \frac{2\pi}{\lambda} \mathbf{t}_N^T \mathbf{a}_m} \right]^T \in \mathbb{C}^{N \times 1}, \quad (1)$$

where λ is the carrier wavelength. Following [16], [17], the channel from the BS to LU m is modeled as Rician fading, i.e.,

$$\mathbf{h}_m(\mathbf{t}) = \sqrt{\frac{\kappa_m \beta_m}{\kappa_m + 1}} \bar{\mathbf{h}}_m(\mathbf{t}) + \sqrt{\frac{\beta_m}{\kappa_m + 1}} \tilde{\mathbf{h}}_m, \quad (2)$$

where β_m is the large-scale fading coefficient, $\kappa_m \geq 0$ is the Rician factor, and $\tilde{\mathbf{h}}_m \sim \mathcal{CN}(\mathbf{0}, \mathbf{I}_N)$ is the NLoS component.¹ Similarly, the channel from the BS to Eve is given by

$$\mathbf{g}(\mathbf{t}) = \sqrt{\frac{\kappa_e \beta_e}{\kappa_e + 1}} \bar{\mathbf{g}}(\mathbf{t}) + \sqrt{\frac{\beta_e}{\kappa_e + 1}} \tilde{\mathbf{g}}, \quad (3)$$

¹As in [16], [17], the NLoS components at different antennas are assumed to be independent and identically distributed (i.i.d.), which is reasonable in rich-scattering environments with $D_{\min} \geq \lambda/2$.

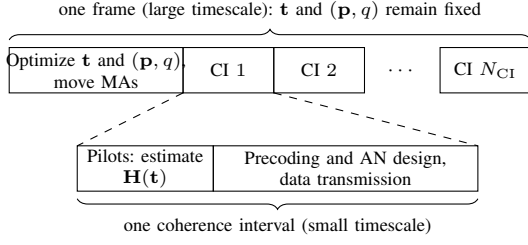


Fig. 2. Illustration of the two-timescale transmission protocol (CI: coherence interval).

where β_e , κ_e , and $\tilde{\mathbf{g}} \sim \mathcal{CN}(\mathbf{0}, \mathbf{I}_N)$ are defined similarly to those in (2), and $\tilde{\mathbf{g}}(\mathbf{t})$ is given by (1) with $\mathbf{a}_m$ replaced by $\mathbf{a}_e \triangleq [\cos \theta_e \sin \phi_e, \sin \theta_e]^T$, where θ_e and ϕ_e are the elevation and azimuth AoDs of Eve. The NLoS components $\tilde{\mathbf{h}}_1, \dots, \tilde{\mathbf{h}}_M$ and $\tilde{\mathbf{g}}$ are mutually independent. We define $\tilde{\mathbf{H}}(\mathbf{t}) \triangleq [\tilde{\mathbf{h}}_1(\mathbf{t}), \dots, \tilde{\mathbf{h}}_M(\mathbf{t})] \in \mathbb{C}^{N \times M}$, $\tilde{\mathbf{H}}(\mathbf{t}) \triangleq [\tilde{\mathbf{h}}_1(\mathbf{t}), \dots, \tilde{\mathbf{h}}_M(\mathbf{t})]$, $\tilde{\mathbf{H}} \triangleq [\tilde{\mathbf{h}}_1, \dots, \tilde{\mathbf{h}}_M]$, and $\mathbf{K} \triangleq \text{diag}(\kappa_1, \dots, \kappa_M)$. Note that the antenna positions affect the channels only through the LoS components $\tilde{\mathbf{H}}(\mathbf{t})$ and $\tilde{\mathbf{g}}(\mathbf{t})$.

B. Two-Timescale Transmission Protocol

Different from the instantaneous CSI over the entire region $\mathcal{C}$, whose acquisition requires the MAs to traverse the region [16], [17], the statistical CSI, i.e., $\{\beta_m, \kappa_m, \theta_m, \phi_m\}_{m \in \mathcal{M}}$ of the LUs and $\{\beta_e, \kappa_e, \theta_e, \phi_e\}$ of Eve, is determined by the large-scale fading and the LoS geometry, which vary much more slowly than the small-scale fading. Following [13], [15], we assume that the statistical CSI of Eve is known at the BS, which can be obtained, e.g., from the prior knowledge of the location of Eve, by sensing its direction [15], or from its historical signals if it is an untrusted node. The impact of errors in the AoD of Eve is evaluated in Section V.

Accordingly, we adopt the two-timescale protocol in Fig. 2. The time is divided into frames, each of which consists of $N_{\text{CI}} \gg 1$ CIs over which the statistical CSI remains constant. At the beginning of each frame, the BS optimizes the antenna positions $\mathbf{t}$ and the long-term transmission parameters based on the statistical CSI, and moves the MAs accordingly. Then, in each CI, the BS estimates $\mathbf{H}(\mathbf{t})$ via uplink pilots by exploiting the channel reciprocity of time-division duplexing, and designs the precoding and AN. For the purpose of exposition, $\mathbf{H}(\mathbf{t})$ is assumed to be perfectly estimated, whereas $\tilde{\mathbf{g}}$ is never available at the BS. Hence, the MAs are moved only once per frame, and the CSI of the LUs is required only at the current antenna positions.

C. Signal Model and Ergodic Secrecy Rate

In each CI, the BS transmits the confidential symbols $s_m \sim \mathcal{CN}(0, 1)$, $m \in \mathcal{M}$, together with AN, i.e.,

$$\mathbf{x} = \sum_{m=1}^M \mathbf{w}_m s_m + \mathbf{z}, \quad (4)$$

where $\mathbf{w}_m \in \mathbb{C}^{N \times 1}$ is the precoding vector for LU m , and $\mathbf{z} \sim \mathcal{CN}(\mathbf{0}, \Sigma)$ is the AN with covariance matrix $\Sigma \succeq \mathbf{0}$, which

is independent of the mutually independent symbols $\{s_m\}$. Under the two-timescale protocol, $\{\mathbf{w}_m\}$ and Σ are functions of the instantaneous CSI of the LUs, which are denoted by $\mathbf{w}_m(\mathbf{H})$ and $\Sigma(\mathbf{H})$ when necessary.

The signal received at LU m is $y_m = \mathbf{h}_m^H(\mathbf{t})\mathbf{x} + n_m$, where $n_m \sim \mathcal{CN}(0, \sigma_m^2)$ is the additive white Gaussian noise (AWGN). The resulting signal-to-interference-plus-noise ratio (SINR) of LU m is

$$\gamma_m = \frac{|\mathbf{h}_m^H \mathbf{w}_m|^2}{\sum_{i \in \mathcal{M} \setminus \{m\}} |\mathbf{h}_m^H \mathbf{w}_i|^2 + \mathbf{h}_m^H \Sigma \mathbf{h}_m + \sigma_m^2}, \quad (5)$$

where the dependence of the channels on $\mathbf{t}$ is omitted for brevity. The signal received at Eve is $y_e = \mathbf{g}^H(\mathbf{t})\mathbf{x} + n_e$ with $n_e \sim \mathcal{CN}(0, \sigma_e^2)$. Since the receiver of a passive Eve is unknown to the BS, we consider the worst case where Eve can cancel the signals of the other LUs when intercepting s_m , e.g., via multiuser detection [20], which yields a secrecy guarantee regardless of the decoding capability of Eve. The SINR of Eve for s_m is then

$$\gamma_{e,m} = \frac{|\mathbf{g}^H \mathbf{w}_m|^2}{\mathbf{g}^H \Sigma \mathbf{g} + \sigma_e^2}, \quad (6)$$

which upper-bounds that of an Eve treating the signals of the other LUs as interference. The latter case is also evaluated in Section V.

Since the instantaneous CSI of Eve is unknown, the confidential message of each LU is encoded over the CIs of a frame, which experience independent realizations of the NLoS components. By coding over these fading realizations, the following ergodic secrecy rate is achievable for LU m [20], [21]:

$$R_m^{\text{sec}} \triangleq [\bar{R}_m - \bar{R}_{e,m}]^+, \quad (7)$$

where

$$\bar{R}_m \triangleq \mathbb{E}\{\log_2(1 + \gamma_m)\}, \quad \bar{R}_{e,m} \triangleq \mathbb{E}\{\log_2(1 + \gamma_{e,m})\} \quad (8)$$

denote the ergodic rate of LU m and the ergodic wiretap rate of Eve for s_m , respectively, and the expectations are taken over $\tilde{\mathbf{H}}$ and $\tilde{\mathbf{g}}$ for a given $\mathbf{t}$. The ESSR is defined as $\sum_{m=1}^M R_m^{\text{sec}}$.

D. Problem Formulation

We aim to maximize the ESSR by jointly optimizing the long-term antenna positions and the short-term precoding and AN design policies. The optimization problem is formulated as

$$(P1) : \max_{\mathbf{t}, \{\mathbf{w}_m(\cdot)\}, \Sigma(\cdot)} \sum_{m=1}^M R_m^{\text{sec}} \quad (9a)$$

$$\text{s.t.} \quad \sum_{m=1}^M \|\mathbf{w}_m(\mathbf{H})\|^2 + \text{tr}(\Sigma(\mathbf{H})) \leq P_{\text{tot}}, \quad (9b)$$

$$\Sigma(\mathbf{H}) \succeq \mathbf{0}, \quad \forall \mathbf{H}, \quad (9c)$$

$$\mathbf{t}_n \in \mathcal{C}, \quad \forall n \in \mathcal{N}, \quad (9d)$$

$$\|\mathbf{t}_n - \mathbf{t}_i\| \geq D_{\min}, \quad \forall n \neq i. \quad (9d)$$

Here, (9b) limits the transmit power to P_{tot} in every CI, (9c) confines the MAs to the movable region, and (9d) enforces the minimum inter-MA distance. Note that $\mathbf{t}$ is shared by all the

CIs of a frame, whereas $\{\mathbf{w}_m(\cdot)\}$ and $\Sigma(\cdot)$ are small-timescale policies.

Problem (P1) is intractable due to three reasons. First, the short-timescale variables are policies, i.e., mappings from the instantaneous CSI of the LUs to the precoders and the AN covariance matrix, which cannot be optimized in each CI separately, since $\bar{R}_{e,m}$ is averaged over all the CIs. Second, the ergodic rates in (8) have no closed-form expressions. Third, $\mathbf{t}$ is coupled with the policies, and (9d) is non-convex. Stochastic optimization based on channel samples [18] is applicable, but its complexity increases with the number of samples, and it provides little insight into the role of the MAs. To address these challenges, we first adopt ZF precoding and null-space AN with a long-term power allocation and derive closed-form approximations of the ergodic rates in Section III, which transform (P1) into a deterministic problem of the long-term variables. Then, we solve this problem by AO in Section IV.

III. TWO-TIMESCALE ZF-AN DESIGN AND ERGODIC SECRECY RATE ANALYSIS

In this section, we first specify the short-timescale ZF precoding and null-space AN. Building on this design, we then derive closed-form approximations of the ergodic rates of the LUs and Eve, which yield an approximate ESSR that depends only on the statistical CSI.

A. Short-Timescale ZF Precoding and Null-Space AN

Given $\mathbf{t}$ and $\mathbf{H} = \mathbf{H}(\mathbf{t})$ in a CI, the BS adopts ZF precoding with a long-term power allocation, i.e., $\mathbf{w}_m = \sqrt{p_m} \bar{\mathbf{w}}_m$, where $p_m \geq 0$ is the power allocated to LU m , and

$$\bar{\mathbf{w}}_m \triangleq \frac{\mathbf{H}(\mathbf{H}^H \mathbf{H})^{-1} \mathbf{e}_m}{\|\mathbf{H}(\mathbf{H}^H \mathbf{H})^{-1} \mathbf{e}_m\|} \quad (10)$$

is the normalized ZF precoding direction of LU m , which satisfies $\mathbf{h}_i^H \bar{\mathbf{w}}_m = 0$, $\forall i \neq m$. Meanwhile, the AN is isotropically distributed in the null space of $\mathbf{H}^H$ [5], [20], i.e.,

$$\Sigma = q \mathbf{P}_\mathbf{H}^\perp, \quad \mathbf{P}_\mathbf{H}^\perp \triangleq \mathbf{I}_N - \mathbf{H}(\mathbf{H}^H \mathbf{H})^{-1} \mathbf{H}^H, \quad (11)$$

where $q \geq 0$ denotes the AN power per dimension. Since $\mathbf{P}_\mathbf{H}^\perp$ is the orthogonal projection onto the null space of $\mathbf{H}^H$, whose dimension is $N - M$ with probability one, we have $\mathbf{H}^H \mathbf{P}_\mathbf{H}^\perp = \mathbf{0}$ and $\text{tr}(\mathbf{P}_\mathbf{H}^\perp) = N - M$. Hence, the AN causes no interference to the LUs, and the transmit power in each CI is $\sum_{m=1}^M p_m + (N - M)q$. The power allocation $(\mathbf{p}, q)$, where $\mathbf{p} \triangleq [p_1, \dots, p_M]^T$, is optimized based on the statistical CSI and remains fixed within a frame, whereas $\{\bar{\mathbf{w}}_m\}$ and $\mathbf{P}_\mathbf{H}^\perp$ are updated in each CI. With this design, the SINRs in (5) and (6) reduce to

$$\gamma_m = \frac{p_m |\mathbf{h}_m^H \bar{\mathbf{w}}_m|^2}{\sigma_m^2} = \frac{p_m}{\sigma_m^2 [(\mathbf{H}^H \mathbf{H})^{-1}]_{m,m}}, \quad (12)$$

$$\gamma_{e,m} = \frac{p_m |\mathbf{g}^H \bar{\mathbf{w}}_m|^2}{q \mathbf{g}^H \mathbf{P}_\mathbf{H}^\perp \mathbf{g} + \sigma_e^2}, \quad (13)$$

respectively, where the second equality in (12) holds since $\mathbf{h}_m^H \mathbf{H}(\mathbf{H}^H \mathbf{H})^{-1} \mathbf{e}_m = 1$ and $\|\mathbf{H}(\mathbf{H}^H \mathbf{H})^{-1} \mathbf{e}_m\|^2 = [(\mathbf{H}^H \mathbf{H})^{-1}]_{m,m}$.

The adopted ZF-AN design requires only the instantaneous CSI of the LUs, eliminates both the inter-user interference and the AN at the LUs, and solves no optimization problem in each CI, since $(\mathbf{p}, q)$ is adapted only to the statistical CSI. The equal power allocation in [16] and the information-AN power splitting with equal power among the LUs are special cases of this power allocation. With the ZF-AN design, (P1) is transformed into

$$(P2) : \max_{\mathbf{t}, \mathbf{p}, q} \sum_{m=1}^M R_m^{\text{sec}}(\mathbf{t}, p_m, q) \quad (14a)$$

$$\text{s.t.} \quad \sum_{m=1}^M p_m + (N - M)q \leq P_{\text{tot}}, \quad (14b)$$

$$p_m \geq 0, \quad \forall m \in \mathcal{M}, \quad q \geq 0, \quad (14c)$$

$$(9c), (9d),$$

where $R_m^{\text{sec}}(\mathbf{t}, p_m, q)$ denotes (7) with the SINRs in (12) and (13). Here, (14b) guarantees (9b) in every CI, and all the variables of (P2) are long-term variables. However, the ergodic rates in (P2) still have no closed-form expressions, which are approximated in the sequel.

B. Ergodic Rate of the LUs

Define $\mathbf{B} \triangleq \text{diag}(\frac{\beta_1}{\kappa_1+1}, \dots, \frac{\beta_M}{\kappa_M+1})$ and $\mathbf{L}(\mathbf{t}) \triangleq \bar{\mathbf{H}}(\mathbf{t}) \mathbf{K}^{1/2}$. From (2), the channel matrix of the LUs can be written as

$$\mathbf{H}(\mathbf{t}) = (\mathbf{L}(\mathbf{t}) + \tilde{\mathbf{H}}) \mathbf{B}^{1/2}, \quad (15)$$

where $\mathbf{L}(\mathbf{t}) + \tilde{\mathbf{H}}$ is the normalized channel matrix, whose LoS component is weighted by the square roots of the Rician factors. Accordingly, we define

$$\Phi(\mathbf{t}) \triangleq \mathbf{L}^H(\mathbf{t}) \mathbf{L}(\mathbf{t}) + N \mathbf{I}_M, \quad \Omega(\mathbf{t}) \triangleq \Phi^{-1}(\mathbf{t}), \quad (16)$$

where $\Phi(\mathbf{t}) = \mathbb{E}\{(\mathbf{L}(\mathbf{t}) + \tilde{\mathbf{H}})^H (\mathbf{L}(\mathbf{t}) + \tilde{\mathbf{H}})\}$ is the correlation matrix of the normalized channels of the LUs.

Lemma 1: For given $\mathbf{t}$ and p_m , the ergodic rate of LU m is approximated by [16], [19]

$$\bar{R}_m \approx \tilde{R}_m(\mathbf{t}, p_m) \triangleq \log_2(1 + \eta_m(\mathbf{t}) p_m), \quad (17)$$

where $\eta_m(\mathbf{t}) \triangleq \zeta_m(N - M)/(N\omega_m(\mathbf{t}))$ is the effective channel gain of LU m , $\zeta_m \triangleq \beta_m/((\kappa_m + 1)\sigma_m^2)$, and $\omega_m(\mathbf{t}) \triangleq [\Omega(\mathbf{t})]_{m,m}$.

Proof: By applying Jensen's inequality to the convex function $\log_2(1 + 1/x)$ of $x > 0$, we have $\bar{R}_m \geq \log_2(1 + p_m/(\sigma_m^2 \mathbb{E}\{[(\mathbf{H}^H \mathbf{H})^{-1}]_{m,m}\}))$. Following [19], the noncentral Wishart matrix $(\mathbf{L} + \tilde{\mathbf{H}})^H (\mathbf{L} + \tilde{\mathbf{H}})$ is approximated by a central Wishart matrix with N degrees of freedom and the same mean $\Phi(\mathbf{t})$, whose inverse has the mean $N\Omega(\mathbf{t})/(N - M)$. Together with (15), this yields $\mathbb{E}\{[(\mathbf{H}^H \mathbf{H})^{-1}]_{m,m}\} \approx N(\kappa_m + 1)\omega_m(\mathbf{t})/(\beta_m(N - M))$, which leads to (17). ■

Lemma 1 is equivalent to the approximation in [16, Sec. IV-A], since the normalized channel correlation matrix used therein equals $\frac{1}{N}(\mathbf{K} + \mathbf{I}_M)^{-1/2} \Phi(\mathbf{t}) (\mathbf{K} + \mathbf{I}_M)^{-1/2}$. It reveals that the antenna positions affect the ergodic rate of LU m only through $\omega_m(\mathbf{t})$. Since $\Phi(\mathbf{t}) \succeq N \mathbf{I}_M$ and $[\Omega(\mathbf{t})]_{m,m} \geq 1/[\Phi(\mathbf{t})]_{m,m} = 1/(N(\kappa_m + 1))$ [22], we have

$$\zeta_m(N - M) \leq \eta_m(\mathbf{t}) \leq \zeta_m(N - M)(\kappa_m + 1), \quad (18)$$

where the upper bound is achieved if $\bar{\mathbf{h}}_m^H(\mathbf{t})\bar{\mathbf{h}}_i(\mathbf{t}) = 0$ for all $i \neq m$ with $\kappa_i > 0$. Hence, $1/(N\omega_m(\mathbf{t})) \in [1, \kappa_m + 1]$ is the LoS gain of LU m that survives the ZF precoding, which can be improved by decorrelating the LoS components of the LUs via the antenna positions.

Remark 1 (Accuracy of Lemma 1): Lemma 1 is a lower-bound-type approximation, which is exact for Rayleigh fading, i.e., $\mathbf{K} = \mathbf{0}$, where (17) reduces to the well-known lower bound $\log_2(1 + \zeta_m p_m(N - M))$ [23]. For large Rician factors, the degrees-of-freedom loss $(N - M)/N$ also applies to the deterministic LoS component, and the gap between (17) and $\bar{R}_m$ approaches $\log_2(N/(N - M))$ at high SNR as $\kappa_i \rightarrow \infty$, $\forall i \in \mathcal{M}$. Since this gap depends mainly on the Rician factors and $N/(N - M)$, and only weakly on $\mathbf{t}$ (see Section V), we adopt (17) due to its simplicity [16], [17] and focus on the ergodic wiretap rate of Eve, for which no existing result is available.

C. Ergodic Wiretap Rate of Eve

Different from γ_m , the SINR of Eve in (13) depends on the random ZF direction $\bar{\mathbf{w}}_m$ and the random AN subspace, both of which are induced by $\mathbf{H}$. To handle this dependence, we first characterize the information leakage and the AN received by Eve conditioned on $\mathbf{H}$ in the following lemma.

Lemma 2: Conditioned on $\mathbf{H}$, the information leakage $X_m \triangleq |\mathbf{g}^H \bar{\mathbf{w}}_m|^2$ and the AN power $Y \triangleq \mathbf{g}^H \mathbf{P}_{\bar{\mathbf{H}}}^\perp \mathbf{g}$ received by Eve are independent, and their conditional means and variances are given by

$$\begin{aligned} \mathbb{E}\{X_m|\mathbf{H}\} &= \tilde{\beta}_e(\kappa_e x_m + 1), \\ \text{Var}\{X_m|\mathbf{H}\} &= \tilde{\beta}_e^2(2\kappa_e x_m + 1), \end{aligned} \quad (19)$$

$$\begin{aligned} \mathbb{E}\{Y|\mathbf{H}\} &= \tilde{\beta}_e(\kappa_e y + N - M), \\ \text{Var}\{Y|\mathbf{H}\} &= \tilde{\beta}_e^2(2\kappa_e y + N - M), \end{aligned} \quad (20)$$

where $\tilde{\beta}_e \triangleq \beta_e/(\kappa_e + 1)$, $x_m \triangleq |\bar{\mathbf{g}}^H \bar{\mathbf{w}}_m|^2$, and $y \triangleq \bar{\mathbf{g}}^H \mathbf{P}_{\bar{\mathbf{H}}}^\perp \bar{\mathbf{g}}$.

Proof: Let $\mathbf{U} \triangleq [\mathbf{u}_1, \dots, \mathbf{u}_{N-M}]$ be an orthonormal basis of the null space of $\mathbf{H}^H$, so that $\mathbf{P}_{\bar{\mathbf{H}}}^\perp = \mathbf{U}\mathbf{U}^H$ and $Y = \sum_{i=1}^{N-M} |\mathbf{u}_i^H \mathbf{g}|^2$. Since $\bar{\mathbf{w}}_m$ is a unit-norm vector in the column space of $\mathbf{H}$, the matrix $[\bar{\mathbf{w}}_m, \mathbf{U}]$ has orthonormal columns. Since $\bar{\mathbf{g}}$ is independent of $\mathbf{H}$, we have $\mathbf{g}|\mathbf{H} \sim \mathcal{CN}(\sqrt{\kappa_e \tilde{\beta}_e} \bar{\mathbf{g}}, \tilde{\beta}_e \mathbf{I}_N)$ from (3), and thus $\bar{\mathbf{w}}_m^H \mathbf{g}, \mathbf{u}_1^H \mathbf{g}, \dots, \mathbf{u}_{N-M}^H \mathbf{g}$ are independent complex Gaussian random variables with variance $\tilde{\beta}_e$ given $\mathbf{H}$. Hence, $X_m = |\bar{\mathbf{w}}_m^H \mathbf{g}|^2$ and Y are conditionally independent. Finally, (19) and (20) follow from $\mathbb{E}\{|z|^2\} = |\mu|^2 + \sigma^2$ and $\text{Var}\{|z|^2\} = \sigma^4 + 2|\mu|^2\sigma^2$ for $z \sim \mathcal{CN}(\mu, \sigma^2)$, together with $\sum_{i=1}^{N-M} |\mathbf{u}_i^H \bar{\mathbf{g}}|^2 = y$. ■

Lemma 2 reveals that, given the channels of the LUs, the information leakage is determined only by the component of the wiretap channel along the ZF direction, whereas the AN received by Eve is determined only by its component in the null space of $\mathbf{H}^H$. Moreover, the conditional relative variances of X_m and Y are $(2\kappa_e x_m + 1)/(\kappa_e x_m + 1)^2$ and $(2\kappa_e y + N - M)/(\kappa_e y + N - M)^2$, respectively. Hence, X_m concentrates when the LoS leakage $\kappa_e x_m$ is strong, which is the case that dominates the ergodic wiretap rate, and Y concentrates when $N - M$ or $\kappa_e y$ is large. This motivates the approximation

$\mathbb{E}\{\log_2(1 + X/Y)\} \approx \log_2(1 + \mathbb{E}\{X\}/\mathbb{E}\{Y\})$ [19, Lemma 1], which is accurate when X and Y concentrate around their means and asymptotically exact as $\kappa_e, \kappa_m \rightarrow \infty$, $\forall m$. Using Lemma 2 and taking the expectation over $\bar{\mathbf{H}}$, we obtain

$$\bar{R}_{e,m} \approx \log_2 \left(1 + \frac{p_m(\kappa_e \mathbb{E}\{x_m\} + 1)}{q(\kappa_e \mathbb{E}\{y\} + N - M) + \bar{\sigma}_e^2} \right), \quad (21)$$

where $\bar{\sigma}_e^2 \triangleq \sigma_e^2/\tilde{\beta}_e = \sigma_e^2(\kappa_e + 1)/\beta_e$ is the normalized noise power of Eve.

In (21), $\mathbb{E}\{x_m\} = \mathbb{E}\{|\bar{\mathbf{g}}^H \bar{\mathbf{w}}_m|^2\}$ and $\mathbb{E}\{y\} = \mathbb{E}\{\bar{\mathbf{g}}^H \mathbf{P}_{\bar{\mathbf{H}}}^\perp \bar{\mathbf{g}}\}$ are the average LoS energies of Eve along the ZF direction of LU m and in the AN subspace, respectively, which have no closed-form expressions. Since $\bar{\mathbf{w}}_m$ and $\mathbf{P}_{\bar{\mathbf{H}}}^\perp$ are invariant to the scaling of the columns of $\mathbf{H}$, they can be computed based on the normalized channel matrix $\mathbf{L}(\mathbf{t}) + \bar{\mathbf{H}}$ in (15). This motivates the following lemma for a generic noncentral Gaussian matrix.

Lemma 3: Let $\mathbf{F} = \bar{\mathbf{F}} + \mathbf{W} \in \mathbb{C}^{N \times k}$ with $k < N$, where $\bar{\mathbf{F}}$ is deterministic and the entries of $\mathbf{W}$ are i.i.d. $\mathcal{CN}(0, 1)$ random variables. Let $\mathbf{b} \in \mathbb{C}^{N \times 1}$ be a deterministic vector, and define $\Phi_{\mathbf{F}} \triangleq \mathbb{E}\{\mathbf{F}^H \mathbf{F}\} = \bar{\mathbf{F}}^H \bar{\mathbf{F}} + N\mathbf{I}_k$, $\bar{\mathbf{w}}_m(\mathbf{F}) \triangleq \mathbf{F}(\mathbf{F}^H \mathbf{F})^{-1} \mathbf{e}_m / \|\mathbf{F}(\mathbf{F}^H \mathbf{F})^{-1} \mathbf{e}_m\|$, and $\mathbf{P}_{\bar{\mathbf{F}}}^\perp \triangleq \mathbf{I}_N - \mathbf{F}(\mathbf{F}^H \mathbf{F})^{-1} \mathbf{F}^H$. Then, we have

$$\mathbb{E}\{|\mathbf{b}^H \bar{\mathbf{w}}_m(\mathbf{F})|^2\} \approx \frac{|\Phi_{\mathbf{F}}^{-1} \bar{\mathbf{F}}^H \mathbf{b}|_m|^2 + \|\mathbf{b}\|^2 [\Phi_{\mathbf{F}}^{-2}]_{m,m}}{[\Phi_{\mathbf{F}}^{-1}]_{m,m}}, \quad (22)$$

$$\mathbb{E}\{\mathbf{b}^H \mathbf{P}_{\bar{\mathbf{F}}}^\perp \mathbf{b}\} \approx \frac{(N - k)(\|\mathbf{b}\|^2 - \mathbf{b}^H \bar{\mathbf{F}} \Phi_{\mathbf{F}}^{-1} \bar{\mathbf{F}}^H \mathbf{b})}{N - k + N \text{tr}(\Phi_{\mathbf{F}}^{-1})}. \quad (23)$$

Moreover, (22) and (23) are exact for $\bar{\mathbf{F}} = \mathbf{0}$, and are asymptotically exact for $\bar{\mathbf{F}} = \varrho \bar{\mathbf{F}}_0$ as $\varrho \rightarrow \infty$, where $\bar{\mathbf{F}}_0$ has full column rank. The right-hand sides of (22) and (23) lie in $[0, \|\mathbf{b}\|^2]$, as do their left-hand sides.

Proof: See Appendix A. ■

Lemma 3 replaces the random matrix $(\mathbf{F}^H \mathbf{F})^{-1}$ by the inverse of its mean, which is accurate when the LoS component dominates or N is large compared with k . In (23), the NLoS component is further restricted to the part of the space that is not captured by the LoS component, and its share is calibrated by the identity $\text{tr}(\mathbf{P}_{\bar{\mathbf{F}}}^\perp) = N - k$, which holds for every realization of $\mathbf{F}$ (see Appendix A). Applying Lemma 3 to (21) yields the following closed-form approximation.

Proposition 1: Define $\mathbf{v}(\mathbf{t}) \triangleq \mathbf{L}^H(\mathbf{t})\bar{\mathbf{g}}(\mathbf{t}) = \mathbf{K}^{1/2} \bar{\mathbf{H}}^H(\mathbf{t})\bar{\mathbf{g}}(\mathbf{t})$ and

$$c_m(\mathbf{t}) \triangleq \frac{|\Omega(\mathbf{t})\mathbf{v}(\mathbf{t})|_m|^2 + N[\Omega^2(\mathbf{t})]_{m,m}}{\omega_m(\mathbf{t})}, \quad (24)$$

$$d(\mathbf{t}) \triangleq \frac{(N - M)(N - \mathbf{v}^H(\mathbf{t})\Omega(\mathbf{t})\mathbf{v}(\mathbf{t}))}{N - M + N \text{tr}(\Omega(\mathbf{t}))}. \quad (25)$$

Then, the ergodic wiretap rate of Eve for s_m is approximated by

$$\bar{R}_{e,m} \approx \tilde{R}_{e,m}(\mathbf{t}, p_m, q) \triangleq \log_2 \left(1 + \frac{\xi_m(\mathbf{t}) p_m}{q \psi(\mathbf{t}) + \bar{\sigma}_e^2} \right), \quad (26)$$

where $\xi_m(\mathbf{t}) \triangleq \kappa_e c_m(\mathbf{t}) + 1$ and $\psi(\mathbf{t}) \triangleq \kappa_e d(\mathbf{t}) + N - M$.

Proof: From (15), $\bar{\mathbf{w}}_m = \bar{\mathbf{w}}_m(\mathbf{L}(\mathbf{t}) + \tilde{\mathbf{H}})$ and $\mathbf{P}_{\tilde{\mathbf{H}}}^\perp = \mathbf{P}_{\mathbf{L}(\mathbf{t}) + \tilde{\mathbf{H}}}^\perp$. Applying Lemma 3 with $\tilde{\mathbf{F}} = \mathbf{L}(\mathbf{t})$, $\mathbf{W} = \tilde{\mathbf{H}}$, $k = M$, and $\mathbf{b} = \bar{\mathbf{g}}(\mathbf{t})$, for which $\Phi_{\mathbf{F}} = \Phi(\mathbf{t})$, $\tilde{\mathbf{F}}^H \mathbf{b} = \mathbf{v}(\mathbf{t})$, and $\|\mathbf{b}\|^2 = N$, yields $\mathbb{E}\{x_m\} \approx c_m(\mathbf{t})$ and $\mathbb{E}\{y\} \approx d(\mathbf{t})$. Substituting them into (21) completes the proof. ■

Proposition 1 reveals that $c_m(\mathbf{t}) \in [0, N]$ and $d(\mathbf{t}) \in [0, N]$ approximate the LoS energy of Eve along the ZF direction of LU m and in the AN subspace, respectively, so that $\xi_m(\mathbf{t})$ and $\psi(\mathbf{t})$ are the normalized information leakage for LU m and the normalized AN power received by Eve. Different from the existing approximations for ZF precoding, which only involve the diagonal entries of $\Omega(\mathbf{t})$, Proposition 1 captures the correlation between the LoS components of Eve and the LUs via $\mathbf{v}(\mathbf{t})$.

D. Approximate ESSR and Discussions

Based on Lemma 1 and Proposition 1, the ergodic secrecy rate of LU m is approximated by

$$R_m^{\text{sec}} \approx \tilde{R}_m^{\text{sec}}(\mathbf{t}, p_m, q) \triangleq [\tilde{R}_m(\mathbf{t}, p_m) - \tilde{R}_{e,m}(\mathbf{t}, p_m, q)]^+, \quad (27)$$

which depends only on the statistical CSI and is thus suitable for the large-timescale design. The physical insights revealed by (27) are discussed in the following remarks.

Remark 2 (Role of the antenna positions): From (17), (24), and (25), the antenna positions affect the approximate ESSR only through three geometric quantities, i.e., $\omega_m(\mathbf{t})$, $c_m(\mathbf{t})$, and $d(\mathbf{t})$. These quantities depend on $\mathbf{t}$ only via $\Phi(\mathbf{t})$ and $\mathbf{v}(\mathbf{t})$, i.e., via the LoS array correlations $[\tilde{\mathbf{H}}^H(\mathbf{t})\tilde{\mathbf{H}}(\mathbf{t})]_{i,j} = \sum_{n=1}^N e^{j\frac{2\pi}{\lambda}\mathbf{t}_n^T(\mathbf{a}_j - \mathbf{a}_i)}$ among the LUs and $[\tilde{\mathbf{H}}^H(\mathbf{t})\bar{\mathbf{g}}(\mathbf{t})]_i = \sum_{n=1}^N e^{j\frac{2\pi}{\lambda}\mathbf{t}_n^T(\mathbf{a}_e - \mathbf{a}_i)}$ between Eve and the LUs. A smaller $\omega_m(\mathbf{t})$ indicates that the LoS component of LU m is less correlated with those of the other LUs, which yields a larger ZF gain; a smaller $c_m(\mathbf{t})$ indicates less information leakage; and a larger $d(\mathbf{t})$ indicates more AN received by Eve. In particular, if $\mathbf{v}(\mathbf{t}) = \mathbf{0}$, i.e., the LoS component of Eve is orthogonal to those of all the LUs, then $c_m(\mathbf{t})$ reduces to its minimum $N[\Omega^2(\mathbf{t})]_{m,m}/\omega_m(\mathbf{t})$ caused by the NLoS components, and $d(\mathbf{t})$ achieves its maximum $N(N-M)/(N-M+N\text{tr}(\Omega(\mathbf{t})))$ for a given $\Phi(\mathbf{t})$. Hence, by decorrelating the LoS component of Eve from those of the LUs, the MAs simultaneously reduce the information leakage and increase the AN received by Eve, which is difficult for an FPA array when Eve is located in a direction close to that of an LU.

Remark 3 (Special cases): 1) If $\kappa_e = 0$, then $\tilde{R}_{e,m} = \log_2(1 + p_m/(q(N-M) + \bar{\sigma}_e^2))$ does not depend on $\mathbf{t}$, i.e., the statistical CSI of Eve carries no spatial information, and the MAs enhance secrecy only by improving $\omega_m(\mathbf{t})$ as in the rate-oriented design. 2) If $M = 1$, then $\Phi(\mathbf{t}) = N(\kappa_1 + 1)$ is a scalar, and $\eta_1 = \zeta_1(N-1)(\kappa_1 + 1)$ does not depend on $\mathbf{t}$.²

²In fact, for $M = 1$, the exact ergodic rate $\mathbb{E}\{\log_2(1 + p_1\|\mathbf{h}_1\|^2/\sigma_1^2)\}$ does not depend on $\mathbf{t}$, since the distribution of $\|\mathbf{h}_1\|^2$ depends on $\mathbf{h}_1(\mathbf{t})$ only through $\|\mathbf{h}_1(\mathbf{t})\|^2 = N$.

In contrast,

$$c_1(\mathbf{t}) = \frac{\kappa_1|\bar{\mathbf{h}}_1^H(\mathbf{t})\bar{\mathbf{g}}(\mathbf{t})|^2/N + 1}{\kappa_1 + 1}, \quad (28)$$

$$d(\mathbf{t}) = \frac{(N-1)(\kappa_1 + 1)}{(N-1)(\kappa_1 + 1) + 1} \left(N - \frac{\kappa_1|\bar{\mathbf{h}}_1^H(\mathbf{t})\bar{\mathbf{g}}(\mathbf{t})|^2}{N(\kappa_1 + 1)} \right) \quad (29)$$

depend on $\mathbf{t}$ through the LoS correlation $|\bar{\mathbf{h}}_1^H(\mathbf{t})\bar{\mathbf{g}}(\mathbf{t})|^2$ between the LU and Eve. Hence, different from the rate-oriented two-timescale design, where the MAs bring no gain for a single user [16], the MAs enhance the secrecy of a single LU by reducing this correlation.

Remark 4 (High-SNR behavior and AN saving): Consider the information-AN power splitting with equal power among the LUs, i.e., $p_m = \alpha P_{\text{tot}}/M$ and $q = (1-\alpha)P_{\text{tot}}/(N-M)$, where $\alpha \in (0, 1)$ is the power-splitting factor. As $P_{\text{tot}} \rightarrow \infty$, we have

$$\tilde{R}_m - \tilde{R}_{e,m} = \log_2 \frac{\alpha P_{\text{tot}} \eta_m(\mathbf{t})}{M} - \log_2 \left(1 + \frac{\alpha \Theta_m(\mathbf{t})}{1-\alpha} \right) + o(1), \quad (30)$$

where $\Theta_m(\mathbf{t}) \triangleq \frac{(N-M)\xi_m(\mathbf{t})}{M\psi(\mathbf{t})}$ is the ratio of the normalized information leakage to the normalized AN power received by Eve. Hence, the high-SNR slope of the ESSR equals M regardless of $\mathbf{t}$, and the antenna positions affect the ESSR through the constant term. Moreover, the value of α that maximizes the sum of the constant terms in (30) is the root of $\sum_{m=1}^M [\frac{1}{\alpha} - \frac{1}{1-\alpha} - \frac{\Theta_m - 1}{1+\alpha(\Theta_m - 1)}] = 0$, which, for $\Theta_m = \Theta$, $\forall m$, is given by

$$\alpha^* = \frac{1}{1 + \sqrt{\Theta}}. \quad (31)$$

Since the MAs can decrease $c_m(\mathbf{t})$ and increase $d(\mathbf{t})$, they decrease Θ and thus increase α^* , i.e., they reduce the AN power required for secure transmission, which is referred to as the AN-saving effect.

IV. PROPOSED TWO-TIMESCALE OPTIMIZATION ALGORITHM

In this section, we reformulate (P2) based on the approximate ESSR, decompose the resulting problem into a power allocation subproblem and an antenna position subproblem, and solve them alternately.

A. Problem Reformulation

By replacing the ergodic secrecy rates in (P2) with their approximations in (27), we consider the following problem, which depends only on the statistical CSI:

$$(P3): \max_{\mathbf{t}, \mathbf{p}, q} F(\mathbf{t}, \mathbf{p}, q) \triangleq \sum_{m=1}^M \left[\log_2(1 + \eta_m(\mathbf{t})p_m) - \log_2 \left(1 + \frac{\xi_m(\mathbf{t})p_m}{q\psi(\mathbf{t}) + \bar{\sigma}_e^2} \right) \right] \quad (32a)$$

s.t. (9c), (9d), (14b), (14c),

where the operator $[\cdot]^+$ in (27) is dropped without loss of optimality, as shown in the following proposition.

Proposition 2: The optimal value of (P3) equals that of the problem obtained by replacing $F(\mathbf{t}, \mathbf{p}, q)$ in (32a) with

$\sum_{m=1}^M \tilde{R}_m^{\text{sec}}(\mathbf{t}, p_m, q)$. Moreover, at any optimal solution of (P3), every term in the summation of (32a) is nonnegative.

Proof: Since $[x]^+ \geq x$, the optimal value with $[\cdot]^+$ is no smaller than that of (P3). Conversely, for any feasible $(\mathbf{t}, \mathbf{p}, q)$, let $\mathbf{p}'$ be obtained by setting $p'_m = 0$ for all m with $\tilde{R}_m(\mathbf{t}, p_m) < \tilde{R}_{e,m}(\mathbf{t}, p_m, q)$ and $p'_m = p_m$ otherwise. Then, $(\mathbf{t}, \mathbf{p}', q)$ is feasible, and since the m -th term of F depends only on p_m and q for a given $\mathbf{t}$, we have $F(\mathbf{t}, \mathbf{p}', q) = \sum_{m=1}^M \tilde{R}_m^{\text{sec}}(\mathbf{t}, p_m, q)$. Hence, the two optimal values are equal. If some term were negative at an optimal solution of (P3), setting the corresponding p_m to zero would strictly increase F , which is a contradiction. ■

Proposition 2 shows that the non-smooth operator $[\cdot]^+$ can be removed by allowing the BS to switch off the LUs with non-positive secrecy rates, which is enabled by the per-user power allocation. In contrast, under equal power allocation among the LUs, the operator $[\cdot]^+$ must be retained. However, (P3) is still intractable due to the coupling between $\mathbf{t}$ and $(\mathbf{p}, q)$ in the objective function, the non-concavity of F w.r.t. $\mathbf{t}$, and the non-convex constraint (9d). To tackle this problem, we adopt the AO framework, which alternately optimizes $(\mathbf{p}, q)$ for given $\mathbf{t}$ and $\mathbf{t}$ for given $(\mathbf{p}, q)$.

B. Power Allocation Optimization for Given $\mathbf{t}$

For given $\mathbf{t}$, the coefficients η_m , ξ_m , and ψ are constants, and (P3) reduces to

$$(P3.1) : \max_{\mathbf{p}, q} \sum_{m=1}^M f_m(p_m; q) \quad \text{s.t. (14b), (14c),} \quad (33)$$

where

$$f_m(p; q) \triangleq \log_2(1 + \eta_m p) - \log_2(1 + \eta_{e,m}(q)p), \quad (34)$$

and $\eta_{e,m}(q) \triangleq \xi_m/(q\psi + \bar{\sigma}_e^2)$ is the effective channel gain of Eve for LU m , which decreases with q . Problem (P3.1) is non-convex in general due to the coupling between p_m and q . Nevertheless, it is convex w.r.t. $\mathbf{p}$ for a given q , as shown below. The first-order and second-order derivatives of f_m w.r.t. p are given by

$$\frac{\partial f_m}{\partial p} = \frac{\eta_m - \eta_{e,m}}{\ln 2 (1 + \eta_m p)(1 + \eta_{e,m} p)}, \quad (35)$$

$$\frac{\partial^2 f_m}{\partial p^2} = \frac{1}{\ln 2} \left[\frac{\eta_{e,m}^2}{(1 + \eta_{e,m} p)^2} - \frac{\eta_m^2}{(1 + \eta_m p)^2} \right], \quad (36)$$

where the argument q of $\eta_{e,m}$ is omitted. If $\eta_m \leq \eta_{e,m}$, then $f_m(p; q) \leq 0$ for all $p \geq 0$, and thus $p_m = 0$ is optimal. If $\eta_m > \eta_{e,m}$, then $\eta_m(1 + \eta_{e,m} p) - \eta_{e,m}(1 + \eta_m p) = \eta_m - \eta_{e,m} > 0$, i.e., $\frac{\eta_m}{1 + \eta_m p} > \frac{\eta_{e,m}}{1 + \eta_{e,m} p}$, which implies that f_m is strictly increasing and strictly concave in $p \geq 0$. Let $\mathcal{M}_a(q) \triangleq \{m \in \mathcal{M} \mid \eta_m > \eta_{e,m}(q)\}$ denote the set of LUs that can achieve positive secrecy rates, and $P_1(q) \triangleq P_{\text{tot}} - (N - M)q$ the power available for the confidential signals. Then, the optimal $\mathbf{p}$ for a given q is given as follows.

Algorithm 1 Power Allocation for Given $\mathbf{t}$

Input: $\{\eta_m, \xi_m\}_{m \in \mathcal{M}}$, ψ , $\bar{\sigma}_e^2$, P_{tot} , and the grid size Q .

Output: q^* and $\mathbf{p}^*$.

- 1: **for** each q in a uniform grid of Q points over $[0, q_{\max}]$ **do**
- 2: Compute $\eta_{e,m}(q)$ and $\mathcal{M}_a(q)$.
- 3: Find ν by bisection such that $\sum_{m \in \mathcal{M}_a(q)} p_m^*(\nu) = P_1(q)$, where $p_m^*(\nu)$ is given by (37).
- 4: Evaluate $G(q)$.
- 5: **end for**
- 6: Repeat Steps 1–5 over a uniform grid of Q points between the two neighbors of the best grid point, and return the best q and its $\mathbf{p}^*$.

Proposition 3 (Secrecy water-filling): For a given q with $P_1(q) > 0$ and $\mathcal{M}_a(q) \neq \emptyset$, the optimal power allocation is $p_m^* = 0$ for $m \notin \mathcal{M}_a(q)$ and

$$p_m^*(\nu) = \left[\frac{2(\rho_m - 1)}{\sqrt{(\eta_m - \eta_{e,m})^2 + 4\eta_m\eta_{e,m}\rho_m} + \eta_m + \eta_{e,m}} \right]^+ \quad (37)$$

for $m \in \mathcal{M}_a(q)$, where $\rho_m = \rho_m(\nu) \triangleq (\eta_m - \eta_{e,m})/(\nu \ln 2)$, and the water level $\nu > 0$ is the unique solution of $\sum_{m \in \mathcal{M}_a(q)} p_m^*(\nu) = P_1(q)$.

Proof: Since the objective is strictly concave and increasing in $\{p_m\}_{m \in \mathcal{M}_a(q)}$ and Slater's condition holds, the Karush–Kuhn–Tucker (KKT) conditions are necessary and sufficient for optimality, and the power constraint is active at the optimum [24]. Let $\nu > 0$ denote the dual variable associated with the power constraint. The KKT conditions yield $\partial f_m / \partial p = \nu$ if $p_m^* > 0$ and $\partial f_m / \partial p|_{p=0} \leq \nu$ otherwise. From (35), the former is equivalent to $(1 + \eta_m p)(1 + \eta_{e,m} p) = \rho_m(\nu)$, i.e., a quadratic equation in p , whose positive root, written in a numerically stable form, is given by (37). The latter is equivalent to $\rho_m(\nu) \leq 1$, in which case (37) gives $p_m^* = 0$. Finally, since $\sum_{m \in \mathcal{M}_a(q)} p_m^*(\nu)$ is continuous and strictly decreasing in ν over $(0, \max_{m \in \mathcal{M}_a(q)} (\eta_m - \eta_{e,m}) / \ln 2)$, ranging from $+\infty$ to 0, the water level is unique. ■

Proposition 3 reveals that (37) reduces to the classic water-filling solution $[1/(\nu \ln 2) - 1/\eta_m]^+$ when $\eta_{e,m} \rightarrow 0$. Hence, (37) can be regarded as a secrecy water-filling solution, in which the LUs whose effective channel gains do not exceed that of Eve are switched off. Since $p_m^*(\nu)$ is non-increasing in ν , the water level can be efficiently found by bisection.

Next, let $G(q) \triangleq \sum_{m \in \mathcal{M}} f_m(p_m^*; q)$ denote the objective value achieved by the optimal $\mathbf{p}$ for a given q , where $G(q) = 0$ if $\mathcal{M}_a(q) = \emptyset$ or $P_1(q) = 0$. The optimal AN power per dimension is obtained by solving

$$q^* = \arg \max_{0 \leq q \leq q_{\max}} G(q), \quad (38)$$

where $q_{\max} \triangleq P_{\text{tot}}/(N - M)$. Increasing q reduces the effective channel gains of Eve at the cost of less power for the confidential signals. Since $G(q)$ is not necessarily unimodal, (38) is solved by a coarse grid search followed by a fine grid search around the best grid point, as summarized in Algorithm 1.

C. Antenna Position Optimization for Given $(\mathbf{p}, q)$

For given $(\mathbf{p}, q)$, (P3) reduces to

$$(P3.2) : \max_{\mathbf{t}} F(\mathbf{t}) \quad \text{s.t. (9c), (9d),} \quad (39)$$

where $F(\mathbf{t}) \triangleq F(\mathbf{t}, \mathbf{p}, q)$. Problem (P3.2) is challenging for two reasons. First, according to Remark 2, $F(\mathbf{t})$ depends on $\mathbf{t}_n$ through the phases $\frac{2\pi}{\lambda} \mathbf{t}_n^T (\mathbf{a}_j - \mathbf{a}_i)$, $i, j \in \mathcal{M} \cup \{e\}$, which rotate by 2π when $\mathbf{t}_n$ moves by $\lambda/\|\mathbf{a}_j - \mathbf{a}_i\|$ along $\mathbf{a}_j - \mathbf{a}_i$. Hence, $F(\mathbf{t})$ is highly oscillatory and has numerous local maxima within $\mathcal{C}$. Second, (9d) is non-convex. Similar to [16], we optimize the antenna positions in an element-wise manner, i.e., we update $\mathbf{t}_n$ with the other positions fixed, for $n = 1, \dots, N$ in turn. Let $F(\mathbf{t}_n)$ denote F as a function of $\mathbf{t}_n$, with the other positions fixed at their latest values. The subproblem for MA n is given by

$$(P3.2.n) : \max_{\mathbf{t}_n \in \mathcal{C}} F(\mathbf{t}_n) \quad \text{s.t. } \|\mathbf{t}_n - \mathbf{t}_i\| \geq D_{\min}, \quad \forall i \neq n. \quad (40)$$

Although (P3.2.n) has only two real variables, its objective function is multimodal. Hence, a local method, such as gradient ascent or MM, converges to the local maximum closest to the current position, so that the MAs move slowly across $\mathcal{C}$ over the AO iterations (see Section V-C). To overcome this issue, we solve (P3.2.n) in two stages, i.e., a global search that locates the best basin of $F(\mathbf{t}_n)$, followed by an MM refinement within this basin, both of which exploit the following rank-one structure.

1) *Rank-one structure*: Let $\mathbf{l}_n(\mathbf{t}_n) \in \mathbb{C}^{M \times 1}$ denote the conjugate transpose of the n -th row of $\mathbf{L}(\mathbf{t})$, i.e.,

$$\mathbf{l}_n(\mathbf{t}_n) = \mathbf{K}^{1/2} \left[e^{-j\frac{2\pi}{\lambda} \mathbf{t}_n^T \mathbf{a}_1}, \dots, e^{-j\frac{2\pi}{\lambda} \mathbf{t}_n^T \mathbf{a}_M} \right]^T. \quad (41)$$

Since $\mathbf{L}^H \mathbf{L} = \sum_{i=1}^N \mathbf{l}_i \mathbf{l}_i^H$ and $\mathbf{L}^H \bar{\mathbf{g}} = \sum_{i=1}^N \mathbf{l}_i e^{j\frac{2\pi}{\lambda} \mathbf{t}_i^T \mathbf{a}_e}$, we have

$$\Phi(\mathbf{t}) = \Gamma_n + \mathbf{l}_n(\mathbf{t}_n) \mathbf{l}_n^H(\mathbf{t}_n), \quad (42)$$

$$\mathbf{v}(\mathbf{t}) = \mathbf{r}_n + \mathbf{l}_n(\mathbf{t}_n) e^{j\frac{2\pi}{\lambda} \mathbf{t}_n^T \mathbf{a}_e}, \quad (43)$$

where $\Gamma_n \triangleq \sum_{i \neq n} \mathbf{l}_i(\mathbf{t}_i) \mathbf{l}_i^H(\mathbf{t}_i) + N \mathbf{I}_M$ and $\mathbf{r}_n \triangleq \sum_{i \neq n} \mathbf{l}_i(\mathbf{t}_i) e^{j\frac{2\pi}{\lambda} \mathbf{t}_i^T \mathbf{a}_e}$ do not depend on $\mathbf{t}_n$. By the Sherman–Morrison formula [22],

$$\Omega(\mathbf{t}) = \Gamma_n^{-1} - \frac{\Gamma_n^{-1} \mathbf{l}_n(\mathbf{t}_n) \mathbf{l}_n^H(\mathbf{t}_n) \Gamma_n^{-1}}{1 + \mathbf{l}_n^H(\mathbf{t}_n) \Gamma_n^{-1} \mathbf{l}_n(\mathbf{t}_n)}. \quad (44)$$

Hence, for given Γ_n^{-1} and $\mathbf{r}_n$, the quantities ω_m , c_m , d , and thus $F(\mathbf{t}_n)$ can be evaluated at any candidate position with only $\mathcal{O}(M^2)$ operations.

2) *Stage 1 (global search)*: Let $\mathcal{G} \subset \mathcal{C}$ denote a uniform grid over $\mathcal{C}$ with spacing d_g , and $\mathcal{G}_n \triangleq \{\mathbf{t} \in \mathcal{G} \mid \|\mathbf{t} - \mathbf{t}_i\| \geq D_{\min}, \quad \forall i \neq n\}$ the set of grid points that are feasible for (P3.2.n). Based on (44), we evaluate $F(\mathbf{t}_n)$ at all the points in $\mathcal{G}_n \cup \{\mathbf{t}_n\}$, where $\mathbf{t}_n$ is the current position of MA n , and obtain

$$\hat{\mathbf{t}}_n = \arg \max_{\mathbf{t} \in \mathcal{G}_n \cup \{\mathbf{t}_n\}} F(\mathbf{t}), \quad (45)$$

which requires $\mathcal{O}(|\mathcal{G}|M^2)$ operations. Since the current position is included in the candidates, $F(\hat{\mathbf{t}}_n) \geq F(\mathbf{t}_n)$. The

spacing d_g is chosen as a fraction of the shortest period $\lambda/\max_{i,j \in \mathcal{M} \cup \{e\}} \|\mathbf{a}_j - \mathbf{a}_i\|$ of the phases in $F(\mathbf{t}_n)$, so that every basin of $F(\mathbf{t}_n)$ contains grid points, and $\hat{\mathbf{t}}_n$ typically lies in the basin of a globally optimal solution of (P3.2.n).

3) *Stage 2 (MM refinement)*: Starting from $\mathbf{t}_n^{(0)} = \hat{\mathbf{t}}_n$, we refine the position by the MM method [25], where $\mathbf{t}_n^{(k)}$ denotes the position of MA n in the k -th MM iteration. To construct a surrogate function, we derive the gradient of $F(\mathbf{t}_n)$. For $u \in \{x_n, y_n\}$, let $\mathbf{D}_u \triangleq \text{diag}(a_{1,u}, \dots, a_{M,u})$, where $a_{m,x} \triangleq \cos \theta_m \sin \phi_m$ and $a_{m,y} \triangleq \sin \theta_m$ are the entries of $\mathbf{a}_m$, and let $a_{e,u}$ denote the corresponding entry of $\mathbf{a}_e$. From (41)–(43), the derivatives of $\mathbf{l}_n$, $\mathbf{v}$, and Ω w.r.t. u are given by

$$\dot{\mathbf{l}}_n \triangleq \frac{\partial \mathbf{l}_n}{\partial u} = -j \frac{2\pi}{\lambda} \mathbf{D}_u \mathbf{l}_n, \quad (46)$$

$$\dot{\mathbf{v}} \triangleq \frac{\partial \mathbf{v}}{\partial u} = j \frac{2\pi}{\lambda} (a_{e,u} \mathbf{I}_M - \mathbf{D}_u) \mathbf{l}_n e^{j\frac{2\pi}{\lambda} \mathbf{t}_n^T \mathbf{a}_e}, \quad (47)$$

$$\dot{\Omega} \triangleq \frac{\partial \Omega}{\partial u} = -\Omega (\dot{\mathbf{l}}_n \mathbf{l}_n^H + \mathbf{l}_n \dot{\mathbf{l}}_n^H) \Omega, \quad (48)$$

where (48) follows from $\partial \Phi^{-1} = -\Phi^{-1} (\partial \Phi) \Phi^{-1}$. Based on (46)–(48), the derivatives $\dot{\omega}_m \triangleq \partial \omega_m / \partial u$, $\dot{c}_m \triangleq \partial c_m / \partial u$, and $\dot{d} \triangleq \partial d / \partial u$ are derived in closed form in Appendix B. Let $\chi_m \triangleq \xi_m p_m$ and $\Upsilon \triangleq q\psi + \bar{\sigma}_e^2$, whose derivatives are $\dot{\chi}_m = \kappa_e p_m \dot{c}_m$ and $\dot{\Upsilon} = \kappa_e q \dot{d}$, respectively. Moreover, $\dot{\eta}_m = -\eta_m \dot{\omega}_m / \omega_m$. By the chain rule,

$$\frac{\partial F}{\partial u} = \frac{1}{\ln 2} \sum_{m=1}^M \left[\frac{p_m \dot{\eta}_m}{1 + \eta_m p_m} - \frac{\kappa_e p_m \dot{c}_m}{\chi_m + \Upsilon} + \frac{\kappa_e q \chi_m \dot{d}}{\Upsilon (\chi_m + \Upsilon)} \right], \quad (49)$$

and the gradient of F w.r.t. $\mathbf{t}_n$ is $\nabla_{\mathbf{t}_n} F = [\partial F / \partial x_n, \partial F / \partial y_n]^T$.

Since $F(\mathbf{t}_n)$ is twice continuously differentiable and $\mathcal{C}$ is compact, $\nabla_{\mathbf{t}_n} F$ is Lipschitz continuous over $\mathcal{C}$ with some constant $\mathcal{L}_n$. According to the descent lemma [26, Lemma 5.7], for any $\delta_n \geq \mathcal{L}_n$, the following quadratic function is a lower bound of $F(\mathbf{t}_n)$, which is tight at $\mathbf{t}_n^{(k)}$:

$$F(\mathbf{t}_n) \geq \hat{F}_n(\mathbf{t}_n | \mathbf{t}_n^{(k)}) \triangleq F(\mathbf{t}_n^{(k)}) + \nabla_{\mathbf{t}_n} F(\mathbf{t}_n^{(k)})^T (\mathbf{t}_n - \mathbf{t}_n^{(k)}) - \frac{\delta_n}{2} \|\mathbf{t}_n - \mathbf{t}_n^{(k)}\|^2. \quad (50)$$

Since $\mathcal{L}_n$ is difficult to obtain, δ_n is determined by backtracking [26], i.e., starting from $\max\{\delta_0, \delta_n^{\text{pre}}/2\}$, where $\delta_0 > 0$ is a constant and δ_n^{pre} is the previously accepted value, δ_n is doubled until the update below satisfies $F(\mathbf{t}_n^{(k+1)}) \geq \hat{F}_n(\mathbf{t}_n^{(k+1)} | \mathbf{t}_n^{(k)})$, which holds once $\delta_n \geq \mathcal{L}_n$.

To handle the minimum-distance constraints, we note that $\|\mathbf{t}_n - \mathbf{t}_i\|^2$ is convex w.r.t. $\mathbf{t}_n$ and is thus lower-bounded by its first-order Taylor expansion at $\mathbf{t}_n^{(k)}$ [16], i.e.,

$$\|\mathbf{t}_n - \mathbf{t}_i\|^2 \geq \|\Delta_{n,i}^{(k)}\|^2 + 2(\Delta_{n,i}^{(k)})^T (\mathbf{t}_n - \mathbf{t}_n^{(k)}), \quad (51)$$

where $\Delta_{n,i}^{(k)} \triangleq \mathbf{t}_n^{(k)} - \mathbf{t}_i$. Replacing $\|\mathbf{t}_n - \mathbf{t}_i\|^2$ in (9d) with the right-hand side of (51) yields the linear constraint

$$(\Delta_{n,i}^{(k)})^T \mathbf{t}_n \geq (\Delta_{n,i}^{(k)})^T \mathbf{t}_n^{(k)} + \frac{D_{\min}^2 - \|\Delta_{n,i}^{(k)}\|^2}{2}, \quad \forall i \neq n, \quad (52)$$

Algorithm 2 Two-Timescale AO Algorithm for (P3)

Input: Statistical CSI, P_{tot} , $\mathcal{C}$, D_{min} , d_g , δ_0 , ϵ_{in} , I_{in} , ϵ , and I_{max} .

Output: $\mathbf{t}^*$, $\mathbf{p}^*$, and q^* .

- 1: Initialize a feasible $\mathbf{t}^{(0)}$, obtain $(\mathbf{p}^{(0)}, q^{(0)})$ by Algorithm 1, and set $\ell = 0$.
- 2: **repeat**
- 3: **for** $n = 1$ to N **do**
- 4: Stage 1: Compute $\mathbf{\Gamma}_n^{-1}$ and $\mathbf{r}_n$, and obtain $\hat{\mathbf{t}}_n$ by (45).
- 5: Stage 2: Starting from $\mathbf{t}_n^{(0)} = \hat{\mathbf{t}}_n$, update $\mathbf{t}_n$ by (53) with backtracking on δ_n until the fractional increase of F is below ϵ_{in} or I_{in} iterations are reached.
- 6: **end for**
- 7: Obtain $(\mathbf{p}', q')$ by Algorithm 1 for the updated positions $\mathbf{t}^{(\ell+1)}$. Set $(\mathbf{p}^{(\ell+1)}, q^{(\ell+1)}) = (\mathbf{p}', q')$ if this does not decrease F , and $(\mathbf{p}^{(\ell+1)}, q^{(\ell+1)}) = (\mathbf{p}^{(\ell)}, q^{(\ell)})$ otherwise.
- 8: $\ell \leftarrow \ell + 1$.
- 9: **until** the fractional increase of F is below ϵ or $\ell = I_{\text{max}}$

whose feasible set is a convex subset of that of (9d) w.r.t. $\mathbf{t}_n$. Based on (50) and (52), $\mathbf{t}_n$ is updated by solving

$$\mathbf{t}_n^{(k+1)} = \arg \max_{\mathbf{t}_n} \hat{F}_n(\mathbf{t}_n | \mathbf{t}_n^{(k)}) \quad \text{s.t. } \mathbf{t}_n \in \mathcal{C}, \quad (53)$$

By completing the square, (53) is equivalent to projecting $\mathbf{t}_n^{(k)} + \nabla_{\mathbf{t}_n} F(\mathbf{t}_n^{(k)}) / \delta_n$ onto the convex polygon defined by $\mathcal{C}$ and (52). Its optimal solution lies at the projected point itself (if feasible), on an edge, or at a vertex of the polygon, and can thus be obtained exactly by enumerating the projections onto the $N+3$ constraint lines and their pairwise intersections. Since $\mathbf{t}_n^{(k)}$ is feasible for (53), the update satisfies

$$F(\mathbf{t}_n^{(k+1)}) \geq \hat{F}_n(\mathbf{t}_n^{(k+1)} | \mathbf{t}_n^{(k)}) \geq \hat{F}_n(\mathbf{t}_n^{(k)} | \mathbf{t}_n^{(k)}) = F(\mathbf{t}_n^{(k)}), \quad (54)$$

and $\mathbf{t}_n^{(k+1)}$ is feasible for (P3.2.n) due to (51). The MM iterations are repeated until the fractional increase of F is below ϵ_{in} or I_{in} iterations are reached. Hence, both stages never decrease the objective value of (P3.2).

D. Overall Algorithm, Convergence, and Complexity

The overall algorithm is summarized in Algorithm 2. Since (P3) is non-convex, the obtained solution may depend on the initial positions. To mitigate this sensitivity, Algorithm 2 can be executed from multiple initializations, e.g., the FPA array and several random feasible positions, and the best solution is selected.

Convergence: In each iteration of Algorithm 2, neither the global search in (45) nor the MM updates in (54) decrease the objective value of (P3), and the power allocation is updated only if the objective value does not decrease. Hence, the objective value is non-decreasing over the iterations. Moreover, it is upper-bounded, e.g., $F \leq \sum_{m=1}^M \log_2(1 + \zeta_m(N - M)(\kappa_m + 1)P_{\text{tot}})$ according to (18) and $p_m \leq P_{\text{tot}}$. Therefore, the objective value of Algorithm 2 is guaranteed to converge.

Complexity: In each iteration, forming Φ and Ω requires $\mathcal{O}(NM^2 + M^3)$ operations. For each MA, $\mathbf{\Gamma}_n^{-1}$ is obtained

from Ω by a rank-one update. Stage 1 requires $\mathcal{O}(|\mathcal{G}|(M^2 + N))$ operations, including the feasibility check of the grid points, where $|\mathcal{G}| = (\lfloor A/d_g \rfloor + 1)^2$. In Stage 2, the gradient in (49) and each evaluation of F require $\mathcal{O}(M^2)$ operations based on (44), and solving (53) by enumeration requires $\mathcal{O}(N^3)$ operations. Algorithm 1 requires $\mathcal{O}(QI_bM)$ operations, where I_b is the number of bisection iterations. Hence, the complexity of Algorithm 2 is $\mathcal{O}(I_o(NM^2 + M^3 + N|\mathcal{G}|(M^2 + N) + NI_{\text{in}}J(M^2 + N^3) + QI_bM))$, where I_o is the number of AO iterations and J is the average number of backtracking trials. In contrast to stochastic optimization [18], this complexity does not depend on the number of channel realizations. Moreover, since Algorithm 2 depends only on the slowly varying statistical CSI, it is executed only once per frame, whereas the ZF precoding and AN generation in each CI require only $\mathcal{O}(NM^2 + M^3)$ operations.

V. SIMULATION RESULTS AND DISCUSSIONS

In this section, we first verify the accuracy of the closed-form approximations and the convergence of Algorithm 2, and then compare the proposed design with benchmark schemes.

A. Simulation Setup and Benchmarks

Unless otherwise specified, we set $N = 8$, $M = 3$, $A = 4\lambda$, $D_{\text{min}} = \lambda/2$, $\kappa_m = \kappa_e = 10$ dB, $\beta_m = \beta_e = 1$, and $\sigma_m^2 = \sigma_e^2 = \sigma^2$, such that P_{tot}/σ^2 is the transmit SNR, which is 20 dB by default. The elevation and azimuth AoDs of the LUs are uniformly distributed within $[-\pi/3, \pi/3]$. To model a challenging scenario, Eve is located in a direction close to that of LU 1, i.e., $\theta_e = \theta_1 + \Delta \cos \tilde{\phi}$ and $\phi_e = \phi_1 + \Delta \sin \tilde{\phi}$, where $\tilde{\phi}$ is uniformly distributed over $[0, 2\pi)$ and $\Delta = 0.1$ rad is the angular offset. For Algorithm 2, we set $d_g = \lambda/10$, which is less than one third of the shortest phase period in Section IV-C, $\delta_0 = 1$, $\epsilon_{\text{in}} = 10^{-6}$, $I_{\text{in}} = 20$, $\epsilon = 10^{-4}$, $I_{\text{max}} = 50$, $Q = 41$, and 60 bisection iterations, and select the best of three initializations, i.e., a 2×4 uniform planar array (UPA) with $\lambda/2$ spacing and two random feasible positions. All the results are averaged over 100 random realizations of the AoDs shared by all the schemes, and each design is evaluated by Monte Carlo simulations over 5×10^3 channel realizations with the exact ZF precoding and null-space AN, i.e., the closed-form approximations are used only for optimization.

We compare the proposed design with three benchmarks: 1) **Rate-oriented MA:** The antenna positions maximize the approximate ergodic sum rate of the LUs with equal power, $\sum_{m=1}^M \log_2(1 + \eta_m(\mathbf{t})P_{\text{tot}}/M)$, which ignores Eve [16], by the same position update as in Section IV-C with the same initializations. It combines the existing two-timescale MA design with AN and thus reveals the gain of the secrecy-aware positioning enabled by Proposition 1. 2) **AS:** The BS selects N of the $2N$ antennas of a 4×4 UPA with $\lambda/2$ spacing by an exhaustive search over all the $\binom{2N}{N}$ subsets based on the approximate ESSR in (P3), similar to [15], which requires no antenna movement but twice as many antennas and an RF switching network. 3) **FPA:** The BS adopts the 2×4 UPA with $\lambda/2$ spacing. All the schemes adopt Algorithm 1, so that Figs. 5 and 6 isolate the effect of the antenna positions. A

sparse 2×4 UPA spanning $\mathcal{C}$ performs worse than the rate-oriented MA, e.g., 17.75 versus 18.08 bps/Hz at 20 dB, and is omitted from the figures.

B. Accuracy of the Closed-Form Approximations

Fig. 3 evaluates the accuracy of Lemma 1 and Proposition 1 for the FPA and random feasible MA positions, where the power is split equally between the confidential signals and the AN. It is observed that the closed-form ergodic wiretap sum rate agrees well with the Monte Carlo results over the entire ranges of the SNR and the Rician factor, with a maximum error of 0.16 bps/Hz, which is negligible in the NLoS-dominant regime, where Lemma 3 is exact. Moreover, the random MA positions yield a lower wiretap rate than the FPA, since their larger aperture reduces the LoS correlation between Eve and LU 1. As discussed in Remark 1, Lemma 1 underestimates the ergodic rates of the LUs, and the gap increases with the Rician factor, but it is almost the same for the two configurations, e.g., 1.55 and 1.66 bps/Hz at $\kappa = 10$ dB and 20 dB. More importantly, at the designs obtained in Fig. 5(a) for 20 dB, the Monte Carlo ESSR exceeds the approximate ESSR by 1.52–1.76 bps/Hz for all the schemes, so that the relative gains predicted by (27) deviate from the Monte Carlo ones by at most 2.4 percentage points. Hence, the conservative approximation preserves the ranking of the designs, which justifies its use for optimization.

C. Convergence Behaviour of Algorithm 2

Fig. 4 shows the objective value of (P3) versus the number of iterations of Algorithm 2 for $N \in \{8, 12\}$ and $P_{\text{tot}}/\sigma^2 \in \{10, 20\}$ dB, where the UPA initialization (3×4 for $N = 12$) is adopted, the stopping criterion is disabled, and the results are averaged over the 100 realizations of the AoDs. It is observed that, for all the settings, the objective value increases monotonically and becomes stable within about 15 iterations, where 77%–92% of the total improvement is achieved in the first iteration on average, and the averaged objective value attains 95% of its improvement within three iterations. With $\epsilon = 10^{-4}$, Algorithm 2 terminates after 12.5–16.3 iterations on average. This is because the global search in Stage 1 moves each MA directly to the best basin of $F(\mathbf{t}_n)$, so that the positions do not have to traverse the movable region through local updates. In contrast, without the global search, i.e., with only the MM updates, the objective value increases slowly, requires 32 iterations to attain 95% of its improvement, and is still 4.2% lower after 50 iterations, since the MAs are trapped in the local maxima near their initial positions. Moreover, the final objective values obtained from two random initializations differ from that of the UPA initialization by only 0.1% on average, which indicates that Algorithm 2 is insensitive to the initial positions.

D. ESSR and AN Power Versus Transmit SNR

Fig. 5(a) shows that the proposed design achieves the highest ESSR over the entire SNR range. At $P_{\text{tot}}/\sigma^2 = 20$ dB, it improves the ESSR by 14.8%, 20.4%, and 31.2% compared

TABLE I
ESSR (BPS/Hz) UNDER DIFFERENT EAVESDROPPER MODELS

P_{tot}/σ^2 (dB)	Worst-case Eve (6)			TIN Eve (55)		
	10	20	30	10	20	30
Proposed	11.42	20.75	30.64	11.85	21.24	31.14
Rate-oriented MA	9.38	18.08	27.84	10.53	19.55	29.38
AS	8.92	17.24	26.98	9.77	18.58	28.39
FPA	7.88	15.82	25.49	8.70	17.24	27.00
Proposed w/o AN	9.94	13.06	13.85	11.00	18.70	27.14

with the rate-oriented MA, AS, and the FPA, respectively, and outperforms each of them in every one of the 100 AoD realizations, e.g., the paired 95% bootstrap confidence interval of the gain over the rate-oriented MA is [13.6%, 15.9%]. Although the rate-oriented MA exploits the same region, it achieves less than half of the gain of the proposed design over the FPA. This is because the rate-oriented positions only decorrelate the LUs, whereas the secrecy gain stems mainly from the Eve-aware positioning enabled by Proposition 1. AS improves the ESSR over the FPA by only 9.0%, since its candidates are restricted to a $\lambda/2$ -spaced grid with an aperture of $1.5\lambda \times 1.5\lambda$. As the SNR increases, the gains over the FPA and the rate-oriented MA approach about 5.1 and 2.8 bps/Hz, respectively, which agrees with Remark 4: the antenna positions do not change the high-SNR slope of the ESSR but improve its constant term.

Fig. 5(b) shows that the optimized AN power fraction increases with the SNR and saturates at high SNR, which agrees with (31). Moreover, the proposed design requires considerably less AN power than all the benchmarks, e.g., 0.27 versus 0.43, 0.43, and 0.47 for the rate-oriented MA, AS, and the FPA at 20 dB. This verifies the AN-saving effect in Remark 4: by decorrelating the LoS component of Eve from those of the LUs, the proposed positions reduce the information leakage and increase the AN received by Eve, so that more power can be allocated to the confidential signals.

Table I evaluates the designs, all of which are optimized under the worst-case model in (6), also against an Eve that treats the signals of the other LUs as interference (TIN), whose SINR is

$$\gamma_{e,m}^{\text{TIN}} = \frac{p_m |\mathbf{g}^H \bar{\mathbf{w}}_m|^2}{\sum_{i \in \mathcal{M} \setminus \{m\}} p_i |\mathbf{g}^H \bar{\mathbf{w}}_i|^2 + q \mathbf{g}^H \mathbf{P}_{\mathbf{H}}^\perp \mathbf{g} + \sigma_e^2}. \quad (55)$$

Three observations can be made. First, the proposed design achieves the highest ESSR under both models, and its ESSR under the worst-case model is only 2.3% lower than that against the TIN Eve at 20 dB, i.e., the secrecy guarantee of the worst-case model comes at a marginal cost. Second, for the proposed design without AN, i.e., Algorithm 2 with $q = 0$, the ESSR increases with the SNR against the TIN Eve, for which the inter-user interference acts as AN, but saturates at about 14 bps/Hz under the worst-case model [4]. Hence, the AN is indispensable for a secrecy performance that does not rely on the decoding capability of Eve. Third, against the TIN Eve, the proposed design still outperforms the rate-oriented MA and the FPA by 8.7% and 23.2% at 20 dB, respectively, which confirms that the gain is not an artifact of the worst-case assumption.

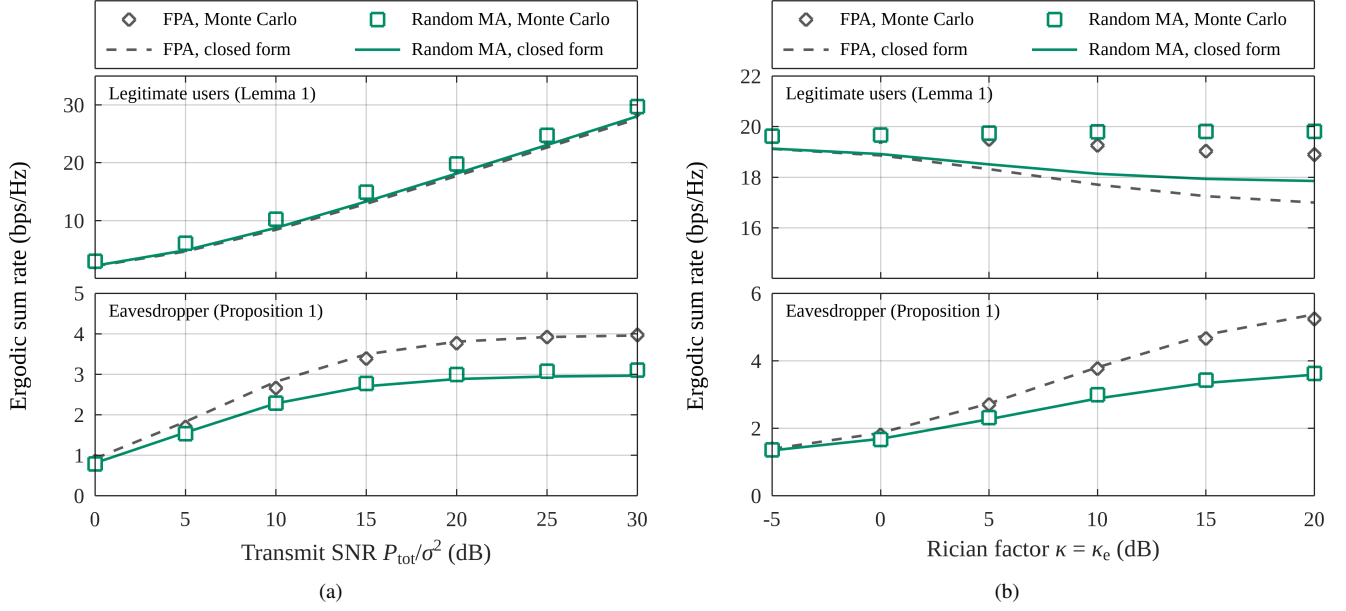


Fig. 3. Ergodic sum rate of the LUs, $\sum_{m=1}^M \bar{R}_m$ (upper panels), and ergodic wiretap sum rate of Eve, $\sum_{m=1}^M \bar{R}_{e,m}$ (lower panels), for the FPA and random MA positions with $p_m = P_{\text{tot}}/(2M)$ and $q = P_{\text{tot}}/(2(N - M))$: (a) versus the transmit SNR with $\kappa_m = \kappa_e = 10$ dB; (b) versus the Rician factor $\kappa = \kappa_m = \kappa_e$ with $P_{\text{tot}}/\sigma^2 = 20$ dB. Markers: Monte Carlo simulations; lines: closed-form approximations in Lemma 1 (LUs) and Proposition 1 (Eve).

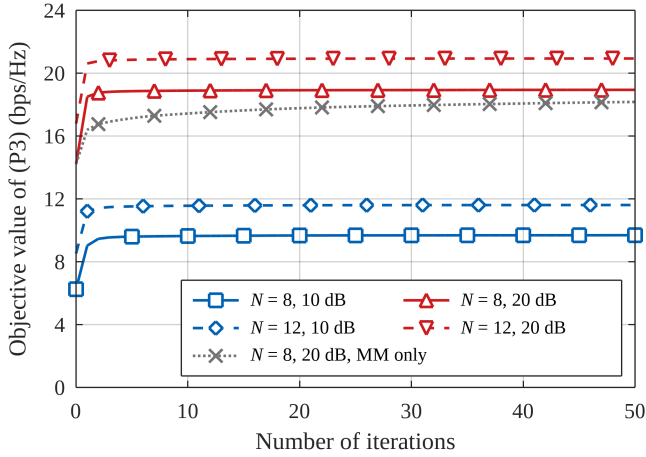


Fig. 4. Convergence behaviour of Algorithm 2 for different numbers of MAs N and transmit SNRs P_{tot}/σ^2 (UPA initialization, averaged over 100 realizations of the AoDs). “MM only”: Algorithm 2 without the global search in Stage 1.

Finally, we examine the robustness to imperfect statistical CSI of Eve, where all the schemes are designed with an AoD of Eve that deviates from the true one by ε in a random direction and are evaluated with the true AoD. For $\varepsilon = 0.01, 0.02$, and 0.05 rad, the ESSR of the proposed design at 20 dB decreases by only 0.1%, 0.5%, and 3.0%, respectively, and its gain over the rate-oriented MA remains above 11%. Even for $\varepsilon = \Delta = 0.1$ rad, the proposed design still outperforms all the benchmarks, e.g., by 4.4% compared with the rate-oriented MA. Hence, the proposed design is robust to moderate errors in the AoD of Eve.

E. Impact of Region Size, Angular Offset, and Number of LUs

Fig. 6 shows the ESSR versus the side length A of the movable region, where the FPA and AS do not depend on A . It is observed that the gain of the proposed design over the FPA increases from 9.2% for $A = 1.5\lambda$, where the region is only slightly larger than the footprint of the UPA, to 38.9% for $A = 6\lambda$. This is because a larger region provides more spatial degrees of freedom for decorrelating the LoS components of Eve and the LUs. In contrast, the rate-oriented MA improves much more slowly with A , i.e., the gain over it increases from 2.7% to 16.6%, which again shows that the region is exploited for secrecy mainly through Eve-aware positioning. For $A = 1.5\lambda$, the candidate array of AS occupies the same region, and AS performs comparably to the proposed design with twice as many antennas, but the gain over AS increases to 27.5% for $A = 6\lambda$. If the $2N$ candidates of AS are instead spread over $\mathcal{C}$ as a 4×4 UPA with spacing $A/3$, AS also benefits from the enlarged aperture and the Eve-aware metric, but the proposed design still outperforms it by 6.1%–8.5% for $A \geq 3\lambda$ with half the number of antennas.

We have also varied the angular offset Δ between Eve and LU 1 from 0.02 to 0.5 rad (the curves are omitted for brevity). For $\Delta = 0.02$ rad, Eve is almost aligned with LU 1, and the proposed design outperforms the rate-oriented MA by only 2.4%, since their LoS components can hardly be decorrelated with an aperture of 4λ . As Δ increases, this gain first increases to 14.8% at $\Delta = 0.1$ rad and then decreases to 10.7% at $\Delta = 0.5$ rad, where Eve becomes separable even with Eve-unaware positions. For a single LU, the rate-oriented MA coincides with the FPA, since the ergodic rate of the LU does not depend on the antenna positions (Remark 3). In contrast, for $\Delta = 0.1$ rad, the proposed design improves the ergodic secrecy rate from 4.22 to 7.48 bps/Hz, i.e., by 77.1%, by reducing the LoS

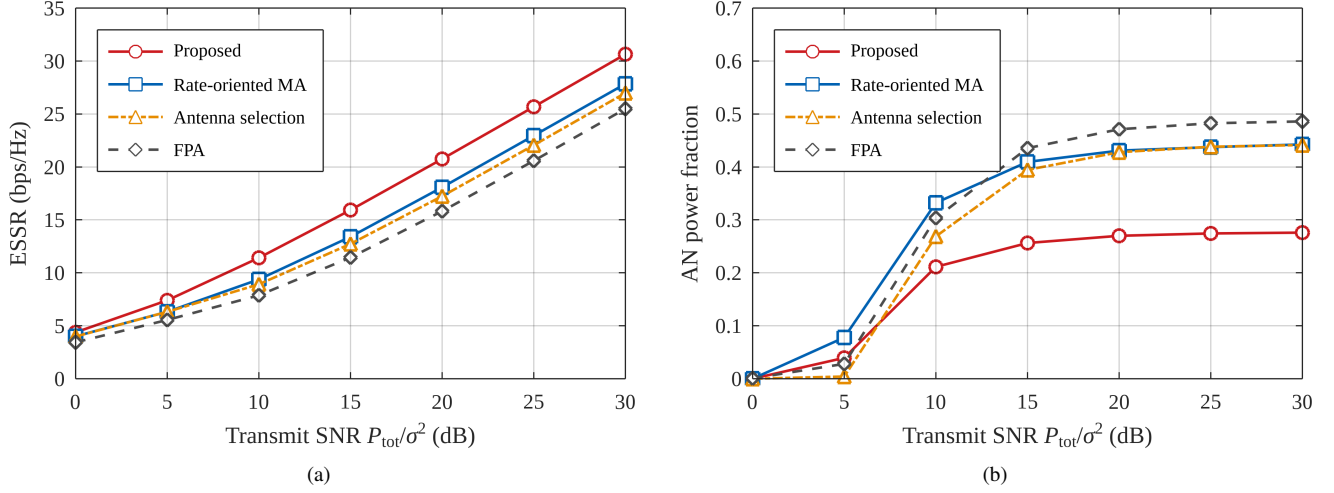


Fig. 5. (a) ESSR and (b) optimized AN power fraction $(N - M)q/P_{\text{tot}}$ versus the transmit SNR P_{tot}/σ^2 .

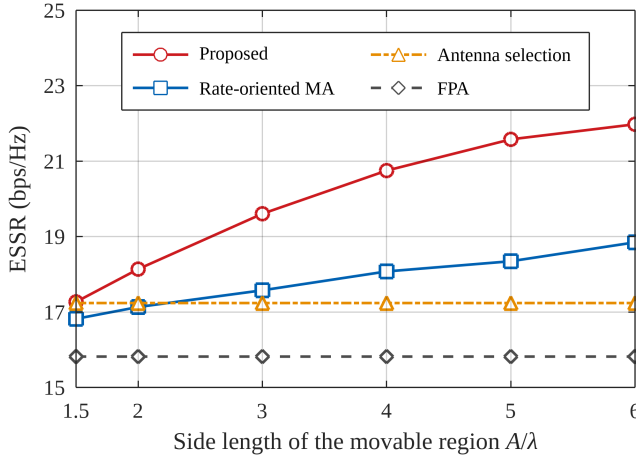


Fig. 6. ESSR versus the side length A of the movable region ($P_{\text{tot}}/\sigma^2 = 20$ dB).

correlation $|\bar{\mathbf{h}}_1^H(\mathbf{t})\bar{\mathbf{g}}(\mathbf{t})|^2$ in (28) and (29), and reduces the AN power fraction from 0.84 to 0.50.

F. Benefit of the Long-Term Power Allocation

Fig. 7 evaluates the benefit of the proposed long-term power allocation, i.e., the secrecy water-filling in Proposition 3 with the optimized AN power. It is compared with two EP benchmarks, i.e., $p_m = \alpha P_{\text{tot}}/M$ and $q = (1 - \alpha)P_{\text{tot}}/(N - M)$, where α is either optimized by a one-dimensional search, similar to [5], or fixed to 0.5, similar to [15]. For all the schemes, the positions are optimized by Algorithm 2, where the objective of the EP benchmarks is $\sum_{m=1}^M \bar{R}_m^{\text{sec}}$, since the LUs cannot be switched off (cf. Proposition 2). Hence, the curves isolate the effect of the power allocation. Three observations can be made. First, for identical LUs, EP with optimized α is nearly optimal, i.e., its ESSR loss is at most 1.2%, where the slightly negative loss at 0 dB is within the Monte Carlo accuracy. This is because the LUs have similar effective channel gains, which are further balanced by the

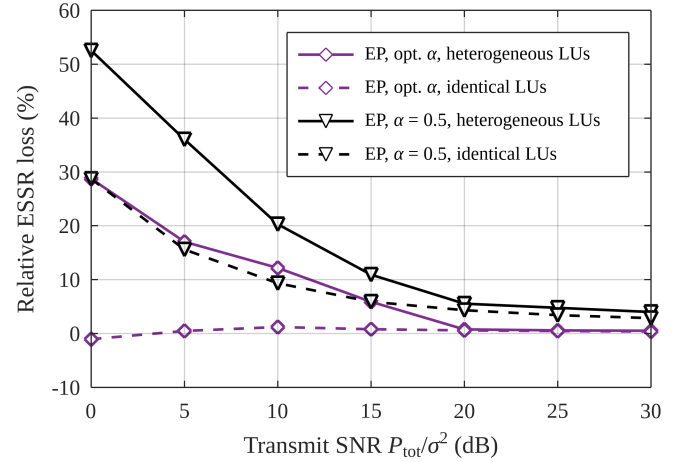


Fig. 7. Relative ESSR loss $100(1 - R_{\text{EP}}/R_{\text{prop}})\%$ of the equal-power (EP) benchmarks w.r.t. the proposed power allocation versus the transmit SNR, for heterogeneous LUs with $[\beta_1, \beta_2, \beta_3] = [0, -10, -20]$ dB and identical LUs. R_{prop} and R_{EP} denote the ESSRs achieved with the proposed and EP power allocations, respectively, where the positions are optimized by Algorithm 2 for each power allocation.

optimized positions, so that the ESSR is mainly determined by the information-AN power splitting (cf. Remark 4). Second, for heterogeneous LUs, the loss is significant at low-to-moderate SNR, e.g., 28.7% and 12.2% at 0 and 10 dB, respectively. This is because the proposed power allocation switches off or throttles the LUs whose effective channel gains are comparable to that of Eve (Proposition 3). Further simulations with $[\beta_1, \beta_2, \beta_3] = [0, -\varsigma/2, -\varsigma]$ dB show that this gain becomes significant for $\varsigma \geq 10$ dB at 0 dB and for $\varsigma \geq 15$ dB at 10 dB, whereas the two allocations perform within about 2% for smaller ς . As the SNR increases, all the LUs achieve positive secrecy rates, and the loss vanishes, similar to conventional water-filling. Third, fixing $\alpha = 0.5$ causes a loss of 2.9%–28.7% for identical LUs and 4.0%–52.5% for heterogeneous LUs, which is most pronounced at low SNR, where little or no AN is needed. Hence, the

information-AN power splitting is the dominant component of the long-term power allocation, whereas the per-user power allocation is beneficial for heterogeneous LUs at low-to-moderate SNR.

VI. CONCLUSION

This paper proposed a two-timescale secure transmission framework for an MA-enabled multiuser downlink system with a passive Eve, where the antenna positions and the long-term power allocation are optimized based on the statistical CSI, and ZF precoding and null-space AN are designed based on the instantaneous CSI of the LUs. We derived a closed-form approximation of the ergodic wiretap rate, which shows that the antenna positions affect the ESSR only through three geometric quantities characterizing the decorrelation among the LUs, the information leakage, and the AN received by Eve. Based on this result, we developed an AO algorithm with secrecy water-filling and a global-search-aided MM position update, which converges within a few iterations. Simulation results demonstrated that the proposed secrecy-aware positioning outperforms the rate-oriented two-timescale MA design, AS, and FPA arrays, enhances the secrecy even for a single LU, and reduces the AN power required for secure transmission. Extending the framework to multiple or multi-antenna eavesdroppers and to imperfect instantaneous CSI of the LUs is an interesting direction for future work.

APPENDIX A PROOF OF LEMMA 3

For brevity, let $\Phi \triangleq \Phi_F$. We first prove (22). Replacing $(\mathbf{F}^H \mathbf{F})^{-1}$ in the unnormalized ZF direction by $\Phi^{-1} = (\mathbb{E}\{\mathbf{F}^H \mathbf{F}\})^{-1}$ yields $\bar{\mathbf{w}}_m(\mathbf{F}) \approx \mathbf{F} \Phi^{-1} \mathbf{e}_m / \|\mathbf{F} \Phi^{-1} \mathbf{e}_m\|$. Then, we approximate the mean of $|\mathbf{b}^H \mathbf{F} \Phi^{-1} \mathbf{e}_m|^2 / \|\mathbf{F} \Phi^{-1} \mathbf{e}_m\|^2$ by the ratio of the means of its numerator and denominator, in the same spirit as [19, Lemma 1]. Since $\mathbb{E}\{\mathbf{W}\} = \mathbf{0}$ and $\mathbb{E}\{\mathbf{W} \mathbf{x} \mathbf{x}^H \mathbf{W}^H\} = \|\mathbf{x}\|^2 \mathbf{I}_N$ for any deterministic $\mathbf{x}$, we have

$$\begin{aligned} \mathbb{E}\{|\mathbf{b}^H \mathbf{F} \Phi^{-1} \mathbf{e}_m|^2\} &= |\mathbf{b}^H \bar{\mathbf{F}} \Phi^{-1} \mathbf{e}_m|^2 + \|\mathbf{b}\|^2 \|\Phi^{-1} \mathbf{e}_m\|^2 \\ &= |[\Phi^{-1} \bar{\mathbf{F}}^H \mathbf{b}]_m|^2 + \|\mathbf{b}\|^2 [\Phi^{-2}]_{m,m}, \end{aligned} \quad (56)$$

$$\mathbb{E}\{\|\mathbf{F} \Phi^{-1} \mathbf{e}_m\|^2\} = \mathbf{e}_m^T \Phi^{-1} \Phi \Phi^{-1} \mathbf{e}_m = [\Phi^{-1}]_{m,m}, \quad (57)$$

which lead to (22). Next, we prove (23). Replacing $(\mathbf{F}^H \mathbf{F})^{-1}$ in $\mathbf{P}_F \triangleq \mathbf{I}_N - \mathbf{P}_F^\perp = \mathbf{F}(\mathbf{F}^H \mathbf{F})^{-1} \mathbf{F}^H$ by Φ^{-1} and using $\mathbb{E}\{\mathbf{W} \Phi^{-1} \mathbf{W}^H\} = \text{tr}(\Phi^{-1}) \mathbf{I}_N$ yield

$$\mathbb{E}\{\mathbf{P}_F\} \approx \mathbf{\Pi} + \text{tr}(\Phi^{-1}) \mathbf{I}_N, \quad \mathbf{\Pi} \triangleq \bar{\mathbf{F}} \Phi^{-1} \bar{\mathbf{F}}^H, \quad (58)$$

where $\mathbf{\Pi}$ and $\text{tr}(\Phi^{-1}) \mathbf{I}_N$ represent the parts of the signal subspace induced by the LoS and NLoS components, respectively. However, (58) lets the NLoS component capture the same fraction $\text{tr}(\Phi^{-1})$ of every direction, including those already captured by the LoS component. Hence, it counts the energy of $\mathbf{b}$ in these directions twice, and $\mathbf{b}^H (\mathbf{I}_N - \mathbf{\Pi} - \text{tr}(\Phi^{-1}) \mathbf{I}_N) \mathbf{b}$ may even become negative when $\mathbf{b}$ is closely aligned with the column space of $\bar{\mathbf{F}}$. To avoid this double counting, we let the NLoS component capture a fraction $\epsilon \in [0, 1]$ of the complement of $\mathbf{\Pi}$ only, i.e.,

$$\mathbb{E}\{\mathbf{P}_F\} \approx \mathbf{\Pi} + \epsilon (\mathbf{I}_N - \mathbf{\Pi}), \quad (59)$$

which satisfies $\mathbf{0} \preceq \mathbf{\Pi} + \epsilon (\mathbf{I}_N - \mathbf{\Pi}) \preceq \mathbf{I}_N$ since $\mathbf{0} \preceq \mathbf{\Pi} \prec \mathbf{I}_N$, as shown below. The fraction ϵ is determined by the identity $\text{tr}(\mathbf{P}_F) = k$, which holds for every realization of $\mathbf{F}$. Since $\text{tr}(\mathbf{\Pi}) = \text{tr}(\Phi^{-1}(\Phi - N \mathbf{I}_k)) = k - N \text{tr}(\Phi^{-1})$, taking the trace of both sides of (59) yields $k - N \text{tr}(\Phi^{-1}) + \epsilon(N - k + N \text{tr}(\Phi^{-1})) = k$, i.e., $\epsilon = N \text{tr}(\Phi^{-1}) / (N - k + N \text{tr}(\Phi^{-1}))$. Therefore, $\mathbb{E}\{\mathbf{P}_F^\perp\} \approx (1 - \epsilon)(\mathbf{I}_N - \mathbf{\Pi})$ with $1 - \epsilon = (N - k) / (N - k + N \text{tr}(\Phi^{-1}))$, which leads to (23).

Range: Let $\mu_1, \dots, \mu_k$ denote the singular values of $\bar{\mathbf{F}}$. The nonzero eigenvalues of $\mathbf{\Pi}$ are $\mu_i^2 / (\mu_i^2 + N) < 1$, and thus $\mathbf{0} \preceq \mathbf{\Pi} \prec \mathbf{I}_N$. Moreover, $\Phi \succeq N \mathbf{I}_k$ implies $0 < \text{tr}(\Phi^{-1}) \leq k/N$, and thus $1 - \epsilon \in [1 - k/N, 1)$. Hence, the right-hand side of (23) lies in $[0, \|\mathbf{b}\|^2]$. For (22), by the Cauchy-Schwarz inequality, $|\Phi^{-1} \bar{\mathbf{F}}^H \mathbf{b}|_m|^2 \leq \|\mathbf{b}\|^2 \mathbf{e}_m^T \Phi^{-1} \bar{\mathbf{F}}^H \bar{\mathbf{F}} \Phi^{-1} \mathbf{e}_m$, and $\|\mathbf{b}\|^2 [\Phi^{-2}]_{m,m} \leq N \|\mathbf{b}\|^2 [\Phi^{-2}]_{m,m}$. Since $\mathbf{e}_m^T \Phi^{-1} (\bar{\mathbf{F}}^H \bar{\mathbf{F}} + N \mathbf{I}_k) \Phi^{-1} \mathbf{e}_m = [\Phi^{-1}]_{m,m}$, the right-hand side of (22) lies in $[0, \|\mathbf{b}\|^2]$.

Exactness for $\bar{\mathbf{F}} = \mathbf{0}$: In this case, $\Phi = N \mathbf{I}_k$ and $\mathbf{\Pi} = \mathbf{0}$, and the right-hand sides of (22) and (23) reduce to $\|\mathbf{b}\|^2/N$ and $\|\mathbf{b}\|^2(1 - k/N)$, respectively. On the other hand, since $\mathbf{W}$ is unitarily invariant, i.e., $\mathbf{Q} \mathbf{W}$ has the same distribution as $\mathbf{W}$ for any unitary matrix $\mathbf{Q}$, and $\bar{\mathbf{w}}_m(\mathbf{Q} \mathbf{W}) = \mathbf{Q} \bar{\mathbf{w}}_m(\mathbf{W})$, $\bar{\mathbf{w}}_m(\mathbf{W})$ is uniformly distributed on the unit sphere, and $\mathbb{E}\{\mathbf{P}_W^\perp\}$ commutes with every unitary matrix. Hence, $\mathbb{E}\{|\mathbf{b}^H \bar{\mathbf{w}}_m(\mathbf{W})|^2\} = \|\mathbf{b}\|^2/N$ and $\mathbb{E}\{\mathbf{P}_W^\perp\} = (1 - k/N) \mathbf{I}_N$, where the latter follows from $\text{tr}(\mathbf{P}_W^\perp) = N - k$.

Asymptotic exactness for $\bar{\mathbf{F}} = \varrho \bar{\mathbf{F}}_0$: As $\varrho \rightarrow \infty$, $\mathbf{F}/\varrho \rightarrow \bar{\mathbf{F}}_0$ almost surely. Since $\bar{\mathbf{w}}_m(\cdot)$ and $\mathbf{P}_\perp^\perp$ are invariant to the scaling of their arguments and continuous at full-column-rank matrices, and $|\mathbf{b}^H \bar{\mathbf{w}}_m(\mathbf{F})|^2$ and $\mathbf{b}^H \mathbf{P}_F^\perp \mathbf{b}$ are bounded by $\|\mathbf{b}\|^2$, the dominated convergence theorem yields $\mathbb{E}\{|\mathbf{b}^H \bar{\mathbf{w}}_m(\mathbf{F})|^2\} \rightarrow |\mathbf{b}^H \bar{\mathbf{w}}_m(\bar{\mathbf{F}}_0)|^2$ and $\mathbb{E}\{\mathbf{b}^H \mathbf{P}_F^\perp \mathbf{b}\} \rightarrow \mathbf{b}^H \mathbf{P}_{\bar{\mathbf{F}}_0}^\perp \mathbf{b}$. On the other hand, $\varrho^2 \Phi^{-1} \rightarrow (\bar{\mathbf{F}}_0^H \bar{\mathbf{F}}_0)^{-1}$, $\mathbf{\Pi} \rightarrow \bar{\mathbf{F}}_0 (\bar{\mathbf{F}}_0^H \bar{\mathbf{F}}_0)^{-1} \bar{\mathbf{F}}_0^H$, and $\text{tr}(\Phi^{-1}) \rightarrow 0$. Hence, the right-hand side of (23) converges to $\mathbf{b}^H \mathbf{P}_{\bar{\mathbf{F}}_0}^\perp \mathbf{b}$, and that of (22) converges to $|\mathbf{b}^H \bar{\mathbf{w}}_m(\bar{\mathbf{F}}_0)|^2$, since the second term in the numerator of (22) decays as $\mathcal{O}(\varrho^{-4})$ whereas the denominator decays as $\mathcal{O}(\varrho^{-2})$. This completes the proof.

APPENDIX B DERIVATIVES OF ω_m , c_m , AND d

Let $o_m \triangleq [\Omega^2]_{m,m}$, $\tau \triangleq \text{tr}(\Omega)$, $\hat{\mathbf{v}} \triangleq \Omega \mathbf{v}$, and $\varpi \triangleq \mathbf{v}^H \Omega \mathbf{v}$. Since Ω is Hermitian, (48) can be rewritten as $\dot{\Omega} = -(\Omega \dot{\mathbf{I}}_n)(\Omega \mathbf{I}_n)^H - (\Omega \mathbf{I}_n)(\Omega \dot{\mathbf{I}}_n)^H$. Hence, the derivatives of ω_m , o_m , and τ w.r.t. $u \in \{x_n, y_n\}$ are given by

$$\dot{\omega}_m = [\dot{\Omega}]_{m,m} = -2\text{Re}\{[\Omega \dot{\mathbf{I}}_n]_m [\Omega \mathbf{I}_n]_m^*\}, \quad (60)$$

$$\begin{aligned} \dot{o}_m &= 2\text{Re}\{[\Omega \dot{\Omega}]_{m,m}\} = -2\text{Re}\{[\Omega^2 \dot{\mathbf{I}}_n]_m [\Omega \mathbf{I}_n]_m^* \\ &\quad + [\Omega^2 \mathbf{I}_n]_m [\Omega \dot{\mathbf{I}}_n]_m^*\}, \end{aligned} \quad (61)$$

$$\dot{\tau} = \text{tr}(\dot{\Omega}) = -2\text{Re}\{\mathbf{1}_n^H \Omega^2 \dot{\mathbf{I}}_n\}, \quad (62)$$

where (61) follows from $[\dot{\Omega} \Omega]_{m,m} = [\Omega \dot{\Omega}]_{m,m}^*$. For c_m in (24), the derivative of $\hat{\mathbf{v}} = \Omega \mathbf{v}$ is

$$\dot{\hat{\mathbf{v}}} = \dot{\Omega} \mathbf{v} + \Omega \dot{\mathbf{v}} = -\Omega \dot{\mathbf{I}}_n (\mathbf{I}_n^H \hat{\mathbf{v}}) - \Omega \mathbf{I}_n (\dot{\mathbf{I}}_n^H \hat{\mathbf{v}}) + \Omega \dot{\mathbf{v}}, \quad (63)$$

and thus

$$\dot{c}_m = \frac{2\text{Re}\{[\tilde{\mathbf{v}}]_m^* [\dot{\tilde{\mathbf{v}}}]_m\} + N\dot{o}_m - c_m\dot{\omega}_m}{\omega_m}. \quad (64)$$

For $d = (1 - \epsilon)(N - \varpi)$ in (25), where $1 - \epsilon = (N - M)/(N - M + N\tau)$ as in Appendix A, since $\mathbf{v}^H \dot{\mathbf{\Omega}} \mathbf{v} = -2\text{Re}\{(\tilde{\mathbf{v}}^H \dot{\mathbf{I}}_n)(\mathbf{I}_n^H \tilde{\mathbf{v}})\}$, we have

$$\dot{\varpi} = 2\text{Re}\{\tilde{\mathbf{v}}^H \dot{\mathbf{v}}\} - 2\text{Re}\{(\tilde{\mathbf{v}}^H \dot{\mathbf{I}}_n)(\mathbf{I}_n^H \tilde{\mathbf{v}})\}, \quad (65)$$

$$\dot{d} = -(1 - \epsilon) \left(\dot{\varpi} + \frac{N(N - \varpi) \dot{\tau}}{N - M + N\tau} \right). \quad (66)$$

Substituting $\dot{\eta}_m = -\eta_m \dot{\omega}_m / \omega_m$, (64), and (66) into (49) yields $\nabla_{\mathbf{t}_n} F$. All the quantities in (60)–(66) are obtained from matrix-vector products with $\mathbf{\Omega}$, which require $\mathcal{O}(M^2)$ operations.

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